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Continuous infusion versus bolus injection of loop diuretics for acute heart failure (Review)

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[Intervention Review]

Continuous infusion versus bolus injection of loop diuretics for acute heart failure

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ABSTRACT

Background

Acute heart failure (AHF) is new onset of, or a sudden worsening of, chronic heart failure characterised by congestion in about 95% of cases or end-organ hypoperfusion in 5% of cases. Treatment often requires urgent escalation of diuretic therapy, mainly through hospitalisation. This Cochrane review evaluated the efficacy of intravenous loop diuretics strategies in treating AHF in individuals with New York Heart Association (NYHA) classification III or IV and fluid overload.

Objectives

To assess the effects of intravenous continuous infusion versus bolus injection of loop diuretics for the initial treatment of acute heart failure in adults.

Search methods

We identified trials through systematic searches of bibliographic databases and in clinical trials registers including CENTRAL, MEDLINE, Embase, CPCI-S on the Web of Science, ClinicalTrials.gov, the World Health Organization (WHO) International Clinical Trials Registry platform (ICTRP), and the European Union Trials register. We conducted reference checking and citation searching, and contacted study authors to identify additional studies. The latest search was performed on 29 February 2024.

Selection criteria

We included randomised controlled trials (RCTs) involving adults with AHF, NYHA classification III or IV, regardless of aetiology or ejection fraction, where trials compared intravenous continuous infusion of loop diuretics with intermittent bolus injection in AHF. We excluded trials with chronic stable heart failure, cardiogenic shock, renal artery stenosis, or end-stage renal disease. Additionally, we excluded

studies combining loop diuretics with hypertonic saline, inotropes, vasoactive medications, or renal replacement therapy and trials where diuretic dosing was protocol-driven to achieve a target urine output, due to confounding factors.

Data collection and analysis

Two review authors independently screened papers for inclusion and reviewed full-texts. Outcomes included weight loss, all-cause mortality, length of hospital stay, readmission following discharge, and occurrence of acute kidney injury. We performed risk of bias assessment and meta-analysis where data permitted and assessed certainty of the evidence.

Main results

The review included seven RCTs, spanning 32 hospitals in seven countries in North America, Europe, and Asia. Data collection ranged from eight months to six years. Following exclusion of participants in subgroups with confounding treatments and different clinical settings, 681 participants were eligible for review. These additional study characteristics, coupled with our strict inclusion and exclusion criteria, improve the applicability of the body of the evidence as they reflect real-world clinical practice.

Meta-analysis was feasible for net weight loss, all-cause mortality, length of hospital stay, readmission, and acute kidney injury. Literature review and narrative analysis explored daily fluid balance; cardiovascular mortality; B-type natriuretic peptide (BNP) change; N-terminal-proBNP change; and adverse incidents such as ototoxicity, hypotension, and electrolyte imbalances. Risk of bias assessment revealed two studies with low overall risk, four with some concerns, and one with high risk. All sensitivity analyses excluded trials at high risk of bias.

Only narrative analysis was conducted for 'daily fluid balance' due to diverse data presentation methods across two studies (169 participants, the evidence was very uncertain about the effect). Results of narrative analysis varied. For instance, one study reported higher daily fluid balance within the first 24 hours in the continuous infusion group compared to the bolus injection group, whereas there was no difference in fluid balance beyond this time point.

Continuous intravenous infusion of loop diuretics may result in mean net weight loss of 0.86 kg more than bolus injection of loop diuretics, but the evidence is very uncertain (mean difference (MD) 0.86 kg, 95% confidence interval (CI) 0.44 to 1.28; 5 trials, 497 participants; $P < 0.001$, $I^2 = 21\%$; very low-certainty evidence). Importantly, sensitivity analysis excluding trials with high risk of bias showed there was insufficient evidence for a difference in bodyweight loss between groups (MD 0.70 kg, 95% CI -0.06 to 1.46; 3 trials, 378 participants; $P = 0.07$, $I^2 = 0\%$).

There may be little to no difference in all-cause mortality between continuous infusion and bolus injection (risk ratio (RR) 1.53, 95% CI 0.81 to 2.90; 5 trials, 530 participants; $P = 0.19$, $I^2 = 4\%$; low-certainty evidence). Despite sensitivity analysis, the direction of the evidence remained unchanged.

No trials measured cardiovascular mortality.

There may be little to no difference in the length of hospital stay between continuous infusion and bolus injection of loop diuretics, but the evidence is very uncertain (MD -1.10 days, 95% CI -4.84 to 2.64; 4 trials, 211 participants; $P = 0.57$, $I^2 = 88\%$; very low-certainty evidence). Sensitivity analysis improved heterogeneity; however, the direction of the evidence remained unchanged.

There may be little to no difference in the readmission to hospital between continuous infusion and bolus injection of loop diuretics (RR 0.85, 95% CI 0.63 to 1.16; 3 trials, 400 participants; $P = 0.31$, $I^2 = 0\%$; low-certainty evidence). Sensitivity analysis continued to show insufficient evidence for a difference in the readmission to hospital between groups.

There may be little to no difference in the occurrence of acute kidney injury as an adverse event between continuous infusion and bolus injection of intravenous loop diuretics (RR 1.02, 95% CI 0.70 to 1.49; 3 trials, 491 participants; $P = 0.92$, $I^2 = 0\%$; low-certainty evidence). Sensitivity analysis continued to show that continuous infusion may make little to no difference on the occurrence of acute kidney injury as an adverse events compared to the bolus injection of intravenous loop diuretics.

Authors' conclusions

Analysis of available data comparing two delivery methods of diuretics in acute heart failure found that the current data are insufficient to show superiority of one strategy intervention over the other. Our findings were based on trials meeting stringent inclusion and exclusion criteria to ensure validity. Despite previous reviews suggesting advantages of continuous infusion over bolus injections, our review found insufficient evidence to support or refute this. However, our review, which excluded trials with clinical confounders and RCTs with high risk of bias, offers the most robust conclusion to date.

PLAIN LANGUAGE SUMMARY

Is the use of continuous infusion of loop diuretics better than bolus injections for acute heart failure?

Key messages

– The use of continuous infusion over bolus injection of loop diuretics may result in little to no difference in all the outcomes we measured.

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- The use of continuous infusion over bolus injection of loop diuretics may result in little to no difference in harmful side effects.
- Our research suggests there may be little to no difference between how loop diuretics are delivered. Future studies should focus less on how they are given and more on how they work for patients outside the hospital setting. We should also take into consideration patient preference and quality of life.

What is acute heart failure?

Acute heart failure is a medical condition that impairs the heart's ability to pump blood effectively throughout the body. As a result, the body perceives a lack of blood and compensates by retaining water to increase blood volume. However, this extra volume does not help and it puts additional strain on the heart, leading to breathlessness, fatigue, and swelling of the legs.

How is acute heart failure treated?

Diuretic medicines (sometimes called water tablets) help by reducing the extra water in the body. Loop diuretics are a specific type of diuretic that act on a specific part of the kidneys and are often administered directly into the bloodstream using a cannula, which is a thin plastic tube that is inserted into a blood vessel. This process may require several days and several doses of diuretics to eliminate the excess water that has accumulated in the body.

What did we want to find out?

We wanted to examine and compare two different approaches to administering the loop diuretics: slow administration over an extended period of time (continuous) and administering the loop diuretic in several individual doses (bolus) to assess if one delivery method was better than the other.

What did we do?

We scrutinised over 3400 journal articles and identified seven trials that met our criteria for analysis. We evaluated these trials for any reported changes in bodyweight, probability of death (mortality), duration of hospitalisation, chance of readmission to hospital after being discharged, and whether one delivery method had any harmful effects on the kidneys compared to the other.

What did we find?

We identified seven studies involving 681 adults aged over 18 years who received treatment in a hospital. The participants' average age ranged from 57 to 82 years. These studies were conducted across 32 hospitals in the USA, Canada, Spain, Sweden, India, Turkey, and China between 2010 and 2021, with data collection durations spanning from eight months to six years. The largest study involved 308 participants, while the smallest included 40 participants.

After our analysis, we found that intervention with either delivery method, continuous or bolus, may make little to no difference in:

- change in bodyweight (evidence from 5 studies in 497 people);
- probability of mortality (evidence from 5 studies in 530 people);
- duration of hospitalisation (evidence from 4 studies in 211 people);
- chance of readmission to hospital after being discharged (evidence from 3 studies in 400 people);
- harmful effects on the kidneys (evidence from 3 studies in 491 people).

Overall, we have little confidence in the evidence because the numbers of studies and participants were small and it is possible that people in the studies were aware of which treatment they were getting.

What are the limitations of the evidence?

Despite reviewing numerous journal articles, only seven studies met the criteria to answer our question. Some of the included studies had sections that were not described well, and made the evidence less reliable.

How up to date is this evidence?

This evidence is up to date to 29 February 2024.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Continuous infusion of intravenous loop diuretics compared to bolus injection of intravenous loop diuretics for adults (aged greater than 18 years) with acute heart failure

Continuous infusion of intravenous loop diuretics compared to bolus injection of intravenous loop diuretics for adults (aged greater than 18 years) with acute heart failure

Patient or population: adults (aged greater than 18 years) with acute heart failure

Setting: hospital

Intervention: continuous infusion of intravenous loop diuretics

Comparison: bolus injection of intravenous loop diuretics

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with bolus injection of intravenous loop diuretics	Risk with continuous infusion of intravenous loop diuretics				
Daily fluid balance in millilitres (measured from admission to discharge)	Narrative analysis reveals varying results. For example, 1 study found higher daily fluid balance within the first 24 hours in the continuous infusion group compared to the bolus injection group, whereas there was no difference in fluid balance beyond this time point at the end of observation period.			169 (2 RCTs)	⊕⊕⊕⊕ Very low ^{a,b,c}	The evidence is very uncertain about the effect of continuous infusion of intravenous loop diuretics on daily fluid balance in millilitres.
Net weight loss (measured from admission to discharge)	The mean net weight loss was 3.46 kg	MD 0.86 kg higher (0.44 higher to 1.28 higher)	-	497 (5 RCTs)	⊕⊕⊕⊕ Very low ^{d,e}	Continuous infusion may result in more weight loss compared to bolus injection, but the evidence is very uncertain. When trials with high risk of bias were excluded there was insufficient evidence for a difference in net weight loss between groups (MD 0.70 kg, 95% CI -0.06 to 1.46; 3 trials, 378 participants; P = 0.07).
All-cause mortality (from admission to discharge)	6 per 100	10 per 100 (5 to 19)	RR 1.53 (0.81 to 2.90)	530 (5 RCTs)	⊕⊕⊕⊕ Low ^{f,g}	There may be little to no difference in all-cause mortality between continuous infusion and bolus injection. When trials with high risk of bias were excluded, findings suggested insufficient evidence for a difference in

all-cause mortality between groups (RR 1.20, 95% CI 0.64 to 2.35; 3 trials, 409 participants; P = 0.55).

Cardiovascular mortality - not measured	This outcome was not measured in any of the included studies		-	-		
Length of hospital stay (from admission to discharge)	The mean length of hospital stay was 8.40 days	MD 1.1 days lower (4.84 lower to 2.64 higher)	-	211 (4 RCTs)	⊕⊕⊕⊕ Very low ^{g,h,i}	There may be little to no difference in length of hospital stay between continuous infusion and bolus injection of loop diuretics, but the evidence is very uncertain. Following exclusion of 1 trial with high risk of bias, there was insufficient evidence for a difference in the length of hospital stay between groups (MD 0.61 days, 95% CI -1.53 to 2.75; 3 trials, 130 participants; P = 0.58).
Readmission to hospital following discharge follow-up: range 30 days to 60 days	32 per 100	27 per 100 (20 to 37)	RR 0.85 (0.63 to 1.16)	400 (3 RCTs)	⊕⊕⊕⊕ Low ^{g,j}	There may be little to no difference in readmission to hospital between continuous infusion and bolus injection. When trials with high risk of bias were excluded, there was insufficient evidence for a difference in the readmission to hospital between groups (RR 0.86, 95% CI 0.63 to 1.19; 2 trials, 368 participants; P = 0.37)
Acute kidney injury (adverse event) (from admission to discharge)	18 per 100	18 per 100 (12 to 26)	RR 1.02 (0.70 to 1.49)	491 (3 RCTs)	⊕⊕⊕⊕ Low ^{g,j}	There may be little to no difference in occurrence of acute kidney injury between continuous infusion and bolus injection. When trials with high risk of bias were excluded, there was insufficient evidence to show that continuous infusion of intravenous loop diuretics may not reduce the occurrence of acute kidney injury as an adverse events compared to the bolus injection of intravenous loop diuretics (RR 1.10, 95% CI 0.72 to 1.68; 2 trials, 410 participants; P = 0.65).

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_439496416283539828.

- ^a Downgraded one level due to study limitations: some concerns for overall risk of bias in one trial.
- ^b Downgraded one level due to inconsistency: one trial presented only a figure and no data for this measure.
- ^c Downgraded two levels due to very serious imprecision in both trials.
- ^d Downgraded two levels due to study limitations: high overall risk of bias in two trials and some concerns overall in another two trials.
- ^e Downgraded one level for imprecision: pooled estimate was based on small sample sizes/event rates and wide 95% CIs.
- ^f Downgraded one level due to study limitations: one trial with high risk of bias did not have any events and the proportion of information from another trial with high risk of bias was insufficient to lower certainty of evidence.
- ^g Downgraded one level for imprecision: pooled estimate was based on small sample sizes/event rates and wide 95% CIs that included the possibility of no effect.
- ^h Downgraded two levels due to study limitations: high overall risk of bias in one trial and some concerns overall in the remaining three trials.
- ⁱ Downgraded one level due to inconsistency: substantial heterogeneity.
- ^j Downgraded one level due to study limitations: the proportion of information obtained from one trial with high risk of bias was not sufficient to lower the certainty of evidence further.

BACKGROUND

Description of the condition

Heart failure is a complex clinical syndrome caused by structural or functional cardiac abnormality, or both, leading to elevated intracardiac pressures at rest or during exertion or reduced cardiac output, or both (Ponikowski 2016). Its global prevalence has increased over time: from 646 per 100,000 people, causing 95 years lived with disability (YLD) per 100,000 people worldwide in 1997, to 842 per 100,000 people, causing 130 YLD per 100,000 people in 2017 (IHME 2018). In the UK, heart failure affected approximately 920,000 people (BHF 2020), caused around 59,000 hospitalisations in 2017/2018, and consumed 2% of National Health Service (NHS) expenditure (NCAP 2019; NICE 2018). The estimated global overall cost of heart failure was 108 billion US dollars in 2012 (Cook 2014). Heart failure is also the most common cause of hospitalisations for people older than 65 years in high-income countries (Braunwald 2015). According to the 2019 national heart failure audit in the UK (using data from 2017/2018), the in-hospital, 30-day, and one-year mortality following admission for heart failure was approximately 10%, 16%, and 32%, respectively (NICOR 2019). In the USA, heart failure was registered as the cause of death in 2,839,205 death certificates in 2018, with a crude mortality rate of 868 per 100,000 population (CDC 2018).

Congestion in heart failure is defined as the accumulation of fluid in the intravascular and interstitial space due to increased cardiac filling pressures and sodium and water retention by the kidneys (Martens 2015). Acute heart failure (AHF) is a new onset or sudden worsening of chronic heart failure, and is characterised by congestion in approximately 95% of cases, or end-organ hypoperfusion in 5% of cases (Nohria 2003; Ponikowski 2016). AHF often necessitates urgent medical attention through escalation of oral diuretics or intravenous diuretics for decongestion on an ambulatory basis or via hospitalisation.

This Cochrane review assessed the use of all intravenous loop diuretics in the treatment of AHF in people with NYHA (New York Heart Association) heart failure classification III or IV with fluid overload. We included both inpatient and ambulatory (hospital or community-based) treatment settings.

Description of the intervention

Loop diuretics are the mainstay of treatment for fluid overload in heart failure. They are recognised as the only medications that can adequately control fluid retention in heart failure within a short period of time (especially via the intravenous route), making them the drug of choice for management of AHF (AHA 2022; Ponikowski 2016). Approximately 90% of people admitted with AHF are treated with intravenous loop diuretics (Llorens 2018; Nieminen 2006).

In this review, we defined the continuous infusion of loop diuretics as intravenous administration of the medication at a fixed or variable rate for more than four hours in a 24-hour period. We defined a bolus injection (intermittent) of loop diuretic as intravenous administration of the medication within a duration of four hours or less (recommended rate of administration of furosemide is that it should not be quicker than 4 mg/minute).

How the intervention might work

Loop diuretics inhibit the reabsorption of sodium and chloride at the loop of Henle, causing removal of sodium and fluid from circulation and therefore reducing congestion, which is the most common feature of AHF. Loop diuretics also have other effects, such as activation of the renin-angiotensin axis and vasodilation. This class of drugs includes furosemide, torsemide, bumetanide, and ethacrynic acid.

The input rate is a major determinant of the pharmacodynamics of loop diuretics. Bolus injection of loop diuretics reaches peak concentration quickly compared to continuous infusion, leading to potent natriuresis and diuresis within 10 minutes of administration. However, this effect tapers with time, and could lead to acute tolerance to the drug through activation of the renin-angiotensin-aldosterone system and sympathetic system or hypoalbuminaemia-induced decrease in response (Castañeda-Hernández 2000; Sjöström 1988). Continuous infusion creates a more constant plasma concentration, which contributes to more 'significant' diuresis within the first 24 hours compared to bolus injection (Llorens 2014; Thomson 2010). Continuous infusion may be better tolerated with less haemodynamic instability and variation in heart rate and blood pressure (Mojtahedzadeh 2004). The constantly lower plasma concentration in continuous infusion of loop diuretics as compared to the bolus injection could potentially reduce the occurrence of its adverse effects, such as ototoxicity (Dormans 1996). However, one previous Cochrane review and two other systematic reviews noted no 'significant' difference in electrolyte disturbance, acute kidney injury, or ototoxicity between the two administration methods (Ng 2018; Salvador 2005; Wu 2014).

Why it is important to do this review

For people with AHF, early initiation of intravenous loop diuretics at sufficient dosage is crucial to reduce in-hospital mortality (Matsue 2017). It has also been postulated that the continuous mode of administration could be more efficacious in people with diuretic resistance, cardio-renal syndrome, or severe right ventricular dysfunction (Ellison 2017). However, there is a lack of clarity and consensus on this preference in international guidelines for heart failure. Both the 2014 National Institute for Health and Care Excellence (NICE) and the 2022 American Heart Association (AHA) guidelines recommended commencing intravenous loop diuretics either as bolus or continuous infusion at doses equal to or greater than their chronic doses for those who are already taking regular diuretics (AHA 2022; NICE 2014). However, in the 2016 European Society of Cardiology heart failure guideline, there was no specification on the recommended mode of intravenous administration of loop diuretics for decompensated heart failure (Ponikowski 2016). In its 2018 guideline, the Cardiological Society of India recommended that clinicians should consider increasing the initial dose of intravenous loop diuretics, switching to continuous infusion or adding diuretics of a different mechanism if the first failed to achieve satisfactory diuresis in decompensated heart failure (Guha 2018). The lack of consensus among guidelines inevitably leads to ongoing debate and considerable confusion among clinicians over the optimal administration methods of intravenous loop diuretics in AHF.

One Cochrane review published in 2005 included eight clinical trials that compared intravenous continuous infusion of loop diuretics

versus bolus injection in 254 participants (Salvador 2005). It noted 'significantly' larger urinary output, shorter hospital stays, and fewer adverse effects in participants receiving continuous infusion of loop diuretics compared to bolus injection. Additionally, one 2018 meta-analysis suggested there was no difference between continuous infusion and bolus injection of furosemide for all-cause mortality, length of hospital stay, and electrolyte disturbance, but that continuous infusion of furosemide was superior to bolus injection with regard to diuretic effect and reduction of B-type natriuretic peptide (BNP) (Ng 2018). However, it is important to note that this meta-analysis included only RCTs that had goal-driven treatment regimens and that the clinical implication of the finding was limited due to small sample size and heterogeneity of the included studies (Ng 2018). Another meta-analysis in 2020 found a higher mean daily urine output and weight loss in the continuous infusion group but included RCTs that had confounding treatments such as vasopressors/inotropes and hypertonic saline and also RCTs that had goal-driven treatment regimens, making it difficult to ascertain the true difference between continuous versus bolus infusion of diuretic in isolation of other confounding factors (Chan 2020).

In the context of ongoing confusion over the choice of administrative method of the intravenous loop diuretics in AHF, and outdated or limited quality of previous systematic reviews on this subject, we aimed to conduct this Cochrane review to include the new trial data using the latest Cochrane methods, in order to provide an updated review of the evidence for the choice of continuous infusion versus bolus injection of loop diuretics in the management of AHF.

OBJECTIVES

To assess the effects of intravenous continuous infusion versus bolus injection of loop diuretics for the initial treatment of acute heart failure in adults.

METHODS

Criteria for considering studies for this review

Types of studies

In line with our study protocol (Zhang 2021), we included RCTs where individual participants were randomised as well as cluster-RCTs. We did not exclude quasi-randomised trials, but rather we ensured we assessed all randomisation processes in our risk of bias analysis (Risk of bias in included studies). We excluded RCTs using a cross-over design as there are inherent limitations in cross-over RCTs that are particularly pertinent within the context of people with AHF. For instance, the carryover effect of loop diuretics poses a significant concern, whereby a patient's previous treatment, such as an infusion regimen, may continue to impact kidney function and electrolyte imbalances even after discontinuation.

We included studies reported as full-text publications. We contacted authors of studies published as abstracts only for additional or unpublished data. We listed the studies as awaiting classification when we were unable to retrieve full data for analysis or risk of bias assessment, or both.

Types of participants

We included adults (18 years of age or older) with AHF, NYHA classification III or IV, from any aetiology or ejection fraction, with or without comorbidity.

For studies that did not explicitly describe the subtypes of heart failure, we regarded participants who were hospitalised or treated in the ambulatory setting with new onset or worsening of heart failure symptoms or signs, or both (such as orthopnoea, dyspnoea, peripheral oedema, pulmonary congestion, and significantly raised B-type natriuretic peptide (BNP) or N-terminal-proBNP (NT-proBNP)) to have AHF. Therefore, we included these people in the review.

We excluded studies that only recruited participants with chronic stable heart failure. If studies had conducted subgroup analysis of participants with AHF alongside those with chronic heart failure, we included the subgroup analysis of AHF participants in the review.

We excluded people with cardiogenic shock.

We excluded people who had end-stage renal disease (ESRD; i.e. an estimated glomerular filtration rate (eGFR) less than 15 mL/minute/1.73 m²) who were on renal replacement therapy, including haemodialysis and peritoneal dialysis. However, we included people who had chronic kidney disease (CKD) (eGFR 15 mL/minute/1.73 m² to 90 mL/minute/1.73 m²).

We excluded studies of participants who presented with flash pulmonary oedema due to renal artery stenosis.

Types of interventions

We included trials that compared intravenous continuous infusion of loop diuretics with intermittent intravenous bolus injection as initial treatment for people with AHF, regardless of dosage and administration rate. We included studies in which continuous infusion of loop diuretics was initiated following a single loading dose of the same medication intravenously, to achieve a peak concentration more rapidly.

We excluded the combination of intravenous loop diuretics with intravenous administration of hypertonic saline, inotropes, or vasoactive medications, and renal replacement therapy, including ultrafiltration, due to confounding factors. However, if the study had subgroup comparisons between intravenous continuous infusion of loop diuretics with bolus injection without the above-mentioned combination, we included the subgroup study.

We included studies in which both groups received intravenous administration of loop diuretics fluid in addition to fluid restriction or regular oral medications for heart failure (including other oral diuretics such as metolazone), or both.

We excluded studies where the dose of diuretics or delivery method was protocol/goal driven to achieve a target urine output.

Types of outcome measures

Reporting one or more of the outcomes listed here in the trial was not an inclusion criterion for this review. Where a published report did not appear to report one of these outcomes, we attempted to access the trial protocol and contacted the trial authors to ascertain whether the outcomes were measured but not reported. Relevant

trials that measured these outcomes but did not report the data at all, or not in a usable format, were included in the review as part of the narrative analysis.

When outcomes were presented at different time points, we used the longest available follow-up.

Primary outcomes

- Daily fluid balance (millilitres)
- Net weight loss (kilograms)
- All-cause mortality
- Cardiovascular mortality

Secondary outcomes

- Length of hospital stay (days)
- Readmission to hospital following discharge for the longest available follow-up stated in the trials
- % change of B-type natriuretic peptide (BNP)
- % change of N-terminal-proBNP (NT-proBNP)
- Adverse events (incidence of ototoxicity, electrolyte imbalance, hypotension, acute kidney injury). We analysed the individual adverse events as separate outcomes.
- Change in six-minute walk test (metres)
- Change in visual analogue scale for dyspnoea (10-cm scale)

Search methods for identification of studies

Electronic searches

We identified trials through systematic searches of the following bibliographic databases.

- Cochrane Central Register of Controlled Trials (CENTRAL; Issue 2, 2024) in the Cochrane Library
- Ovid MEDLINE (1946 to 23 February 2024)
- Ovid Embase (1980 to 2024 week 8)
- Conference Proceedings Citation Index-Science (CPCI-S) on the Web of Science (Clarivate Analytics, 1990 to 29 February 2024)

Detailed original search strategies used for each database and search-terms used can be found in [Appendix 1](#). We applied the Cochrane sensitivity and precision maximising RCT filter ([Lefebvre 2020](#)) to MEDLINE and adaptations of the filter to the other databases.

We also conducted a search of the following online clinical trial registries.

- ClinicalTrials.gov (www.clinicaltrials.gov) (searched 29 February 2024)
- European Union Clinical Trials Register (www.clinicaltrialsregister.eu) (searched 29 February 2024)
- World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) Search Portal (trialsearch.who.int) for ongoing or unpublished trials (searched 23 February 2024)

We imposed no restriction on language of publication or publication status. We did not perform a separate search for adverse effects of interventions used for the treatment of AHF. We considered adverse effects described in included studies only.

To ensure our review did not exclude any recent, relevant, literature and to adhere to Cochrane MECIR standard C37, our latest search dates are listed above. These searches form part of our top-up searches in addition to our original search conducted on September 2021. The top-up searches was performed with two small amendments to the search strategies. First an amended search date-range so as not to duplicate our previous search results, and an updated search term extension. See [Appendix 2](#) for detailed search strategies used for the latest top-up searches.

Searching other resources

We checked reference lists of included studies and any relevant systematic reviews identified for additional references to trials. We also examined any relevant retraction statements and errata for included studies and contacted authors for missing data, published abstracts, conference proceedings, and ongoing trials.

Data collection and analysis

Selection of studies

Teams of two review authors (from EF, AT, JZ, RF, MI) independently screened for inclusion. We coded the titles and abstracts of all potential studies that were identified from the searches as 'retrieve' (eligible or potentially eligible/unclear) or 'do not retrieve'. We discussed any disagreements and another review author (RH) arbitrated if necessary.

We retrieved the full-text study reports or publications. Teams of two review authors (from JZ, SA, EF, AT, MI, JB, TL) independently screened the full-texts, identified studies for inclusion, and identified and recorded reasons for exclusion of the ineligible records. We resolved any disagreements through discussion or, if required, we consulted a third review author (RH). Two review authors (DR, JZ) independently screened the yield from the top-up search. We resolved any disagreements by discussion and, another review author (RH) arbitrated if necessary.

We identified and excluded duplicates before starting and during screening, and collated multiple reports of the same study so that each study, rather than each publication, was the unit of interest in the review. We recorded the selection process in sufficient detail to complete a PRISMA flow diagram and [Characteristics of included studies](#) table ([Liberati 2009](#)).

Data extraction and management

We used a data collection form for all relevant study characteristics, which we piloted on at least one study in the review. Three members of the review author team (JZ, JB, SCC) extracted the following study characteristics from included studies. All studies reviewed were published in English, thus, did not require any formal translation prior to data collection.

- Methods: study design, total duration of study, number of study centres and location, study setting, date of study and publication year of the primary reference.
- Participants: number of people randomised, lost to follow-up or withdrawn, analysed; mean age, age range, gender; diagnostic criteria for acute decompensated heart failure (ADHF), subtypes of ADHF (de novo AHF or deterioration of chronic heart failure), severity of the heart failure (symptoms and signs as adopted by the trials, including NYHA classification), inclusion criteria and

exclusion criteria, presence and percentage of participants with CKD.

- Interventions: intervention, comparison, concomitant medications and excluded medications.
- Outcomes: primary and secondary outcomes specified and collected, and time points reported.
- Notes: funding for trials and conflicts of interest of trial authors.
- For subgroup analysis in: CKD versus normal renal function; de novo ADHF versus deterioration of chronic heart failure; and hospital versus ambulatory treatment.

Teams of two review authors (from JZ, SB, SCC) independently extracted outcome data from included studies. We resolved disagreements by consensus or by involving a third review author (RH). One review author (JZ) transferred data into Review Manager (RevMan 2020). Another review author (DR) double-checked that the data were entered correctly by comparing the data presented in the review with the data extraction form. A fourth review author (CZ) spot-checked study characteristics for accuracy against the trial report.

Assessment of risk of bias in included studies

Four members from the review team (JZ, SB, SCC, CZ) independently assessed risk of bias for each study using the Cochrane RoB 2 tool (RoB 2), outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2020a). We resolved any disagreements by discussion or by involving another review author (RH). Another review author (DR) double-checked the outcomes. We assessed the risk of bias in the trials relating to the individual outcomes of daily fluid balance (millilitre), net weight loss (kilograms), all-cause mortality, cardiovascular mortality (subsequently not assessed as there were no studies with data pertaining to the outcome), length of stay in hospital, readmission to hospital, and occurrence of acute kidney injury according to these domains:

- bias arising from the randomisation process;
- bias due to deviations from intended interventions;
- bias due to missing outcome data;
- bias in measurement of the outcome; and
- bias in selection of the reported result.

We assessed the risk of bias for the outcomes of the included trials that we included in [Summary of findings 1](#). If various outcome measures were noted among different trials for the same outcome, we presented the outcome that was most relevant clinically in the summary of findings table, and assessed its risk of bias. Likewise, we presented the outcome with the longest available follow-up duration as stated by the studies, and assessed the potential risk of bias in the outcome at the specific time.

We were interested in assessing the effects of assignment to the intervention at the baseline, irrespective of whether the intervention allocated was indeed received by the participant (the 'intention to treat' principle).

We used the signalling questions in the RoB 2 tool and rated each domain as 'low risk of bias', 'some concerns', or 'high risk of bias'. We summarised the risk of bias judgements across different studies for each of the domains listed for each outcome. The overall risk of

bias for the result was the least favourable assessment across the domains of bias.

We used the RoB 2 Excel tool to manage the assessment of the bias (www.riskofbias.info). We created visual representation graphs using the Robvis platform (McGuinness 2020). All the consensus risk of bias data results files, with RoB 2 signalling questions and responses, are stored and made available for review (Rasoul 2024).

When considering treatment effects, we considered the risk of bias for the studies that contributed to that outcome.

Measures of treatment effect

We analysed dichotomous data as risk ratios (RR) with 95% confidence intervals (CI) and continuous data as mean difference (MD) or standardised mean difference (SMD) with 95% confidence intervals. We used MD when studies used continuous scales of measurement, such as fluid balance or change of bodyweight, and SMD if studies used different scales. We entered data presented as a scale with a consistent direction of effect. We interpreted the effect size of SMD as: less than or equal to 0.2: small or minor; 0.2 to 1.8: medium; equal to or greater than 1.8: large. (Cohen 1988). When the meta-analysis of the outcome of continuous data achieved statistical significance, we aimed to assess the clinical significance of the result by comparing the finding to a minimal clinically important difference (MCID) from the literature (Jakobsen 2014). For the visual analogue scale of dyspnoea, the MCID is a 10.5-mm change on a 10-cm scale in ADHF (Pang 2017). For the six-minute walk test, the MCID is a change of 45 m (Shoemaker 2012). For other continuous outcomes with no validated MCID from literature, the review author team conferred to decide on the MCID when applicable.

We narratively described skewed data reported as medians and interquartile ranges.

None of the trials in our analysis exhibited time-to-event data for re-admissions. In lieu of such data, we delineated the outcome as reported in each respective study. Among the three studies contributing to this analysis, the reported data pertain to the frequency of occurrences, with participants manifesting at least one event during the follow-up period.

Unit of analysis issues

We collected and analysed each outcome at the maximum follow-up for each participant and performed subgroup analysis based on the length of maximum follow-up when applicable. For trials with multiple arms, we combined the relevant treatment arms (e.g. small dose bolus injection and large dose bolus injection) when applicable. To avoid double-counting or omission of participants, we adjusted the calculation of the continuous variates as appropriate; for example, by dividing the denominator in the control arm by two in a three-arm trial.

None of the trials included in our review employed a cluster-randomisation strategy; rather, all trials randomised each individual participant.

Out of all the measured outcomes ([Effects of interventions](#)), the outcome 2.2 ('readmission to hospital following discharge for the longest available follow-up stated in the trials') was the only outcome that could feasibly have re-occurred in the same

participant during a given follow-up period. The other outcomes could be considered as 'non-reoccurring events' and, thus, posed no risk to unit-of-analysis error with re-occurring events. For outcome 2.2, we presented the data as provided in each study. The three studies that presented data on this outcome presented the number of occurrences with participants with at least one event in the follow-up period.

Dealing with missing data

We contacted investigators or study sponsors to verify key study characteristics and obtain missing numerical outcome data where possible (e.g. when a study was reported in an abstract only). Where possible, we used the calculating tool provided by Review Manager to calculate missing standard deviations (SD) using other data from the trial, such as CIs, based on methods outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2020b). Where this was not possible, and the missing data were thought to introduce serious bias, we explored the impact of including such studies in the overall assessment of results through sensitivity analysis.

Assessment of heterogeneity

We inspected forest plots visually to consider the direction and magnitude of effects and the degree of overlap between CIs. We used the I^2 statistic to measure heterogeneity amongst the trials in each analysis but acknowledge that there is substantial uncertainty in the value of the I^2 statistic when there are only a few studies. We also considered the P value from the χ^2 test. If we identified substantial and considerable heterogeneity ($I^2 > 50\%$), we reported it and explored possible causes by prespecified subgroup analysis, as suggested by the *Cochrane Handbook for Systematic Reviews of Interventions* (Deeks 2021).

Assessment of reporting biases

As we could not pool more than 10 trials, we did not create and examine a funnel plot to explore possible small-study biases for the primary outcomes.

Data synthesis

We included all studies for the primary analysis. We undertook meta-analyses only where this is meaningful: that is, if the treatments, participants, and the underlying clinical question were similar enough for pooling to make sense. We had planned to perform sensitivity analyses including only studies with overall low risk of bias, but due to the small number of trials for inclusion meeting those criteria, we performed sensitivity analysis excluding trials with a high overall risk of bias.

We used a random-effects model for meta-analysis, considering the different demographic characteristics and specific intervention among cohorts.

Subgroup analysis and investigation of heterogeneity

We had planned to carry out the following subgroup analyses for any outcomes with substantial heterogeneity. We planned to use the formal test for subgroup differences in Review Manager (RevMan 2020), and to base our interpretation on this.

- Treatment efficiency in participants with CKD (eGFR less than 60 mL/minute/1.73 m²) as opposed to participants without

CKD. Participants with renal insufficiency would require an increasing dose of loop diuretics to achieve diuresis due to altered pharmacokinetics (Shankar 2003).

- Participants with a new diagnosis of ADHF as opposed to acute decompensation of chronic heart failure. Prolonged use of loop diuretics is associated with structural adaptation of the renal tubules, which could contribute to diuretic resistance (Kaissling 1985). AHF is defined as rapid onset or worsening of symptoms or signs of heart failure, or both, requiring urgent evaluation and treatment. It could be a new onset or consequence of acute decompensation of chronic heart failure (Ponikowski 2016).
- Hospital management versus ambulatory or home-based management. As compared to hospital admissions, ambulatory or home-based diuresis for decompensated heart failure has become more accessible, efficient, and safe with potentially favourable outcomes affecting the quality of life of people with heart failure (Ahmed 2021; Graffagnino 2020).

However, no subgroup analysis was performed due to lack of available subgroup data from studies included.

Sensitivity analysis

We carried out sensitivity analysis based on the overall risk of bias. If any studies had missing data, we considered this to introduce serious bias, and we explored the impact of including such studies through sensitivity analysis.

We compared the results for each outcome including all studies versus including only the studies with overall low risk and some concerns of bias.

Summary of findings and assessment of the certainty of the evidence

We created a summary of findings table using the following outcomes: daily fluid balance (millilitres), net weight loss (kilograms), all-cause mortality, cardiovascular mortality, length of hospital stay (days), readmission to hospital following discharge, occurrence of acute kidney injury as an adverse event.

We used the overall RoB 2 judgement to feed into the GRADE assessment. We used the five GRADE considerations (study limitations, consistency of effect, imprecision, indirectness, and publication bias) to assess the certainty of the body of evidence as it related to the studies that contributed data to the meta-analyses for the prespecified outcomes. We used methods and recommendations described in Chapter 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Schünemann 2020), using GRADEpro GDT software (GRADEpro GDT). We justified all decisions to downgrade the certainty of the evidence using footnotes, and we made comments to aid readers' understanding of the review where necessary.

Two review authors (DR, RS) independently judged the certainty of the evidence, and resolved disagreements through discussion or by involving a third review author (RH). We justified, documented, and incorporated judgements into the reporting of results for each outcome.

RESULTS

Description of studies

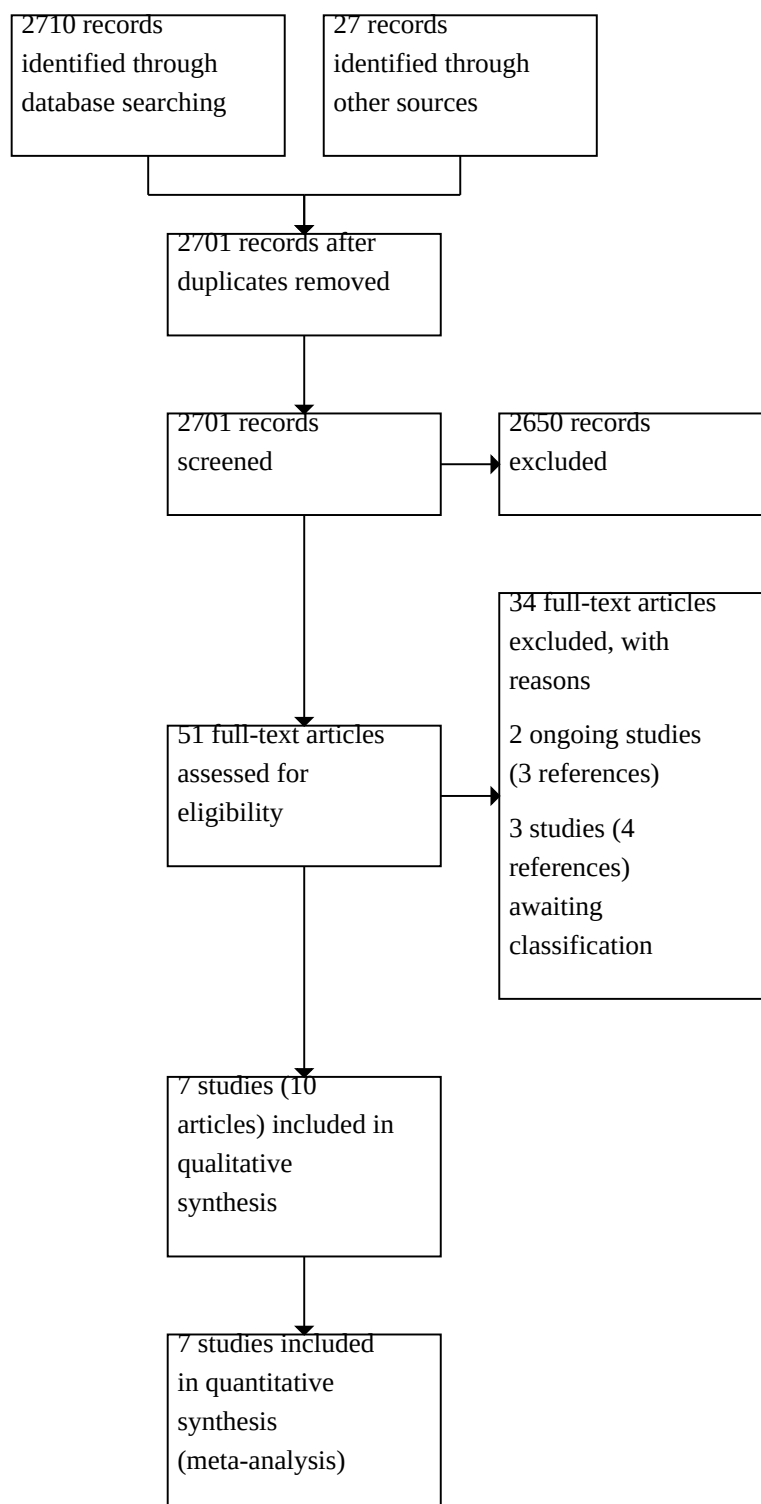
For full details, see [Characteristics of included studies](#); [Characteristics of excluded studies](#); [Characteristics of studies awaiting classification](#); and [Characteristics of ongoing studies](#) tables.

Results of the search

The original electronic search conducted in September 2021 identified 2710 records and 27 records through other sources

([Figure 1](#)). After the removal of duplicates, we screened 2701 records by title and abstract. We assessed 51 full-text records for eligibility. Seven studies were eligible for inclusion in the review ([Allen 2010](#); [Felker 2011](#); [Llorens 2014](#); [Sager 2020](#); [Shah 2014](#); [Yayla 2015](#); [Zheng 2021](#)). We identified two ongoing studies ([NCT03863626](#); [Palazzuoli 2017](#)). One study has no relevant outcome information in the direct comparison between subgroups of participants who received continuous infusions of furosemide versus bolus injections without concomitant dopamine infusions ([Sharma 2018](#)), and two other studies were available solely as abstracts ([Cienki 2009](#); [Ho 2014](#)), thereby we deemed these three studies as awaiting classification.

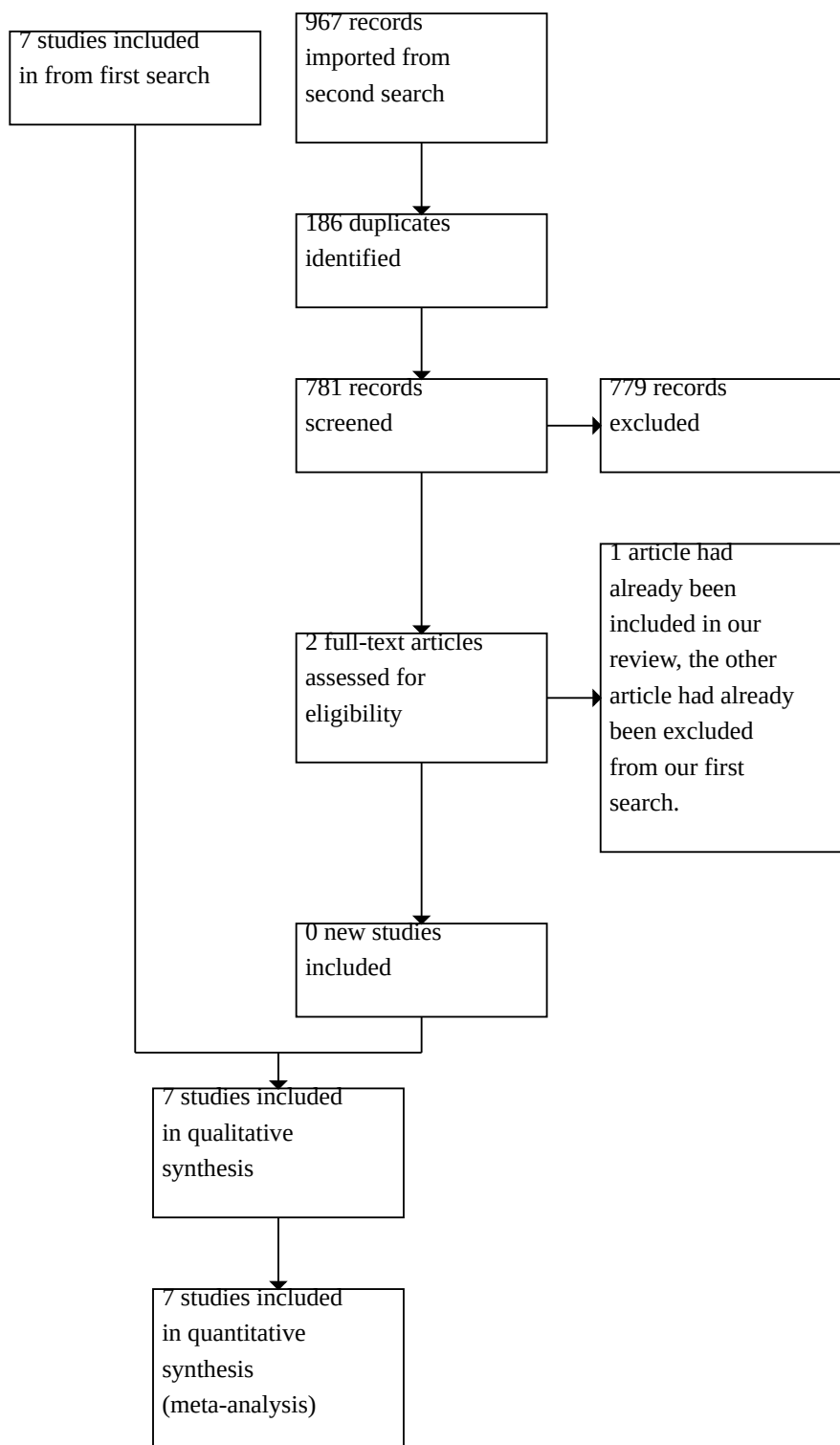
Figure 1. PRISMA flow diagram following the initial search.



The top-up search conducted in February 2024 identified an additional 967 studies. Following removal of duplicates, we screened 781 records by title and abstract. We assessed two records for full-text review and eligibility. One article was already identified and included in the review during the first search ([Zheng 2021](#)), and

the other article was already excluded due to ineligible study design in the first search ([Fudim 2021](#)). Hence, no further studies were included in the review. See [Figure 2](#) for a PRISMA figure including the top-up search.

Figure 2. Flow diagram following the top-up search



In total, we identified 3704 records, screened 3482 records, and performed full-text review on 55 records.

Included studies

This review included seven RCTs with 725 participants (Allen 2010; Felker 2011; Llorens 2014; Sager 2020; Shah 2014; Yayla 2015; Zheng 2021); however, after further exclusion of 30 participants from a subgroup that received concomitant intravenous dopamine (Shah 2014) and 14 participants in a subgroup that received concomitant hypertonic saline infusion alongside intravenous loop diuretics (Yayla 2015), there were 681 participants eligible for inclusion to the review.

We assembled full details of these RCTs in the [Characteristics of included studies](#) table. A narrative description of the characteristics is provided below.

Participants

The included studies were all RCTs in adults aged greater than 18 years. They were performed across 32 centres in the USA (Allen 2010, Felker 2011), Canada (Felker 2011), Spain (Llorens 2014), Sweden (Sager 2020), India (Shah 2014), Turkey (Yayla 2015), and China (Zheng 2021). All enrolled participants had to meet the criteria for decompensation of heart failure by the presence of at least one symptom (from dyspnoea, orthopnoea, or oedema) and one sign (from rales, peripheral oedema, ascites, or pulmonary vascular congestion on chest radiography) necessitating the need for intravenous diuretic therapy. The studies were performed between 2010 and 2021 and collected data over various durations, ranging from eight months to six years. The mean age ranged from 57 years to 82 years.

Interventions

All participants included in the studies were hospitalised in a hospital ward, necessitating intravenous diuretic intervention for a minimum duration of 12 hours. Furosemide was the exclusive intravenous loop diuretic utilised. The 'continuous' infusion cohorts were subjected to intravenous furosemide administration either at a fixed or variable rate for a cumulative period exceeding four hours within a 24-hour time frame. In contrast, the 'bolus' groups received loop diuretic via intravenous administration within a duration of four hours or less.

Outcomes

There was a wide range of prespecified primary endpoints. Allen 2010 measured change in creatinine from admission to hospital day three or discharge, Felker 2011 measured global assessment of symptoms using a visual analogue scale, Llorens 2014 measured urinary output and net diuresis at certain time intervals, Shah 2014 assessed negative fluid balance at 24 hours, and Zheng 2021 assessed freedom from congestion at 72 hours. Sager 2020 and Yayla 2015 did not have any prespecified primary endpoints. Secondary outcomes, safety endpoints, and other measured outcomes are presented in more detail in the [Characteristics of included studies](#) table.

Funding

Five studies did not declare a source of funding (Allen 2010; Llorens 2014; Shah 2014; Yayla 2015; Zheng 2021). Felker 2011 received funding from the National Heart, Lung, and Blood Institute and Sager 2020 received funding from the Swedish Heart-Lung Foundation and the Swedish SUS Fund.

Contact with trial authors

We attempted to contact the corresponding authors of six included studies in order to obtain clarification on study methodology and any potential gaps in data (Allen 2010; Llorens 2014; Sager 2020; Shah 2014; Yayla 2015; Zheng 2021). We received a reply from Sager 2020, helping to clarify their randomisation methodology and concealment of allocation. We did not approach Felker 2011 as we did not require any clarifications on research methodology or missing data.

Excluded studies

We excluded 34 references during the process of the initial full-text screening, and the main reasons for exclusion include 12 studies for ineligible study design (Aaser 1997; Bagatin 1993; ChiCTR1800017270; Dormans 1996; Dormans 1997; Fudim 2021; Lahav 1992; Moon 2012; Pivac 1998; Ruocco 2019; Shah 2012; van Meyel 1993), two studies for ineligible study population (Alvarez 2007; Kramer 1996), and 20 studies for ineligible intervention (EUCTR2013-000866-12-IT; Frea 2020; Kelly 2015; McFarland 1968; NCT00904488; NCT01441245; NCT03592836; NCT03892148; NCT03999216; Palazzuoli 2013; Palazzuoli 2014a; Palazzuoli 2014b; Palazzuoli 2015a; Palazzuoli 2015b; Ragab 2017; Ragab 2018; Schuller 1997; Taema 2018; Thomson 2010; Verel 1964).

During the top-up search, we assessed two studies during the full-text review. One study had already been excluded in our first search (Fudim 2021) and the other study had already been included in our review (Zheng 2021). Thus, the number of excluded trials remained unchanged (37).

See [Characteristics of excluded studies](#) table for full details.

Studies awaiting classification

Three studies are awaiting classification (Cienki 2009; Ho 2014; Sharma 2018).

Ongoing studies

Two studies are ongoing (NCT03863626; Palazzuoli 2017).

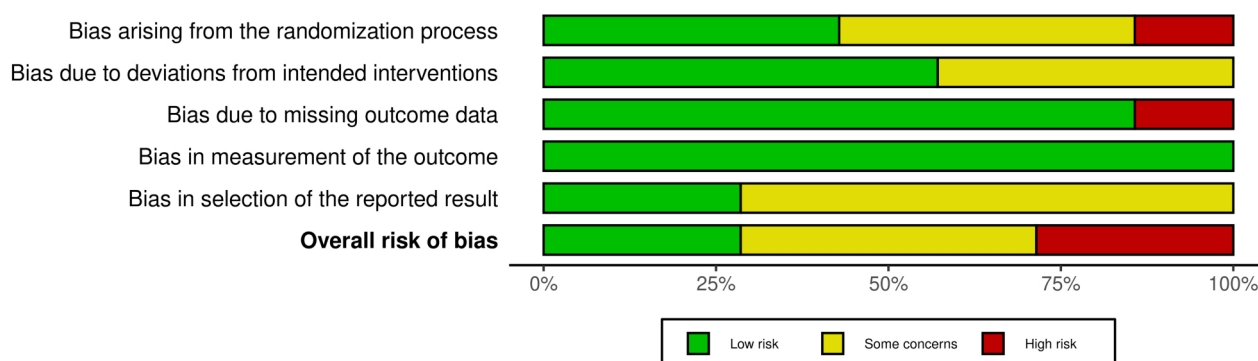
Risk of bias in included studies

Detailed risk of bias assessments including all domain judgements and support for judgement for each relevant outcome measure reported as numerical data are presented as risk of bias tables next to the forest plots in the relevant [Data and analyses](#) section and in the [Risk of bias \(tables\)](#) section.

The risk of bias assessments at outcome level is presented below. We display the summarised risk of bias within the individual studies in [Figure 3](#). The overall risk of bias for a trial is the least favourable assessment score it achieved in any of the domains. We have summarised risk of bias assessment at 'domain' level in [Figure 4](#).

Figure 3. Risk of bias and applicability concerns summary: review authors' consensus judgements about each domain for each included study.

		Risk of bias domains					
		D1	D2	D3	D4	D5	Overall
Study	Allen 2010						
	Felker 2011						
	Llorens 2014						
	Sager 2020						
	Shah 2014						
	Yayla 2015						
	Zheng 2021						
Domains:		Judgement					
D1: Bias arising from the randomization process.		 High					
D2: Bias due to deviations from intended intervention.		 Some concerns					
D3: Bias due to missing outcome data.		 Low					
D4: Bias in measurement of the outcome.							
D5: Bias in selection of the reported result.							

Figure 4. Risk of bias – domain summary overview.

The complete consensus RoB 2 judgements documents with responses to the signalling questions can be found at the Open Science Framework platform (osf.io/nxc53)

Daily fluid balance (millilitres)

This outcome was not included for meta-analysis as explained below ([Effects of interventions](#), 1.1 Daily fluid balance (millilitres)). As we provided a narrative analysis for this outcome, we felt the need to undertake a risk of bias analysis. The consensus judgement document for this can also be found in [Rasoul 2024](#). Two studies were included in our narrative analysis for this outcome, one of which had low risk of bias ([Llorens 2014](#)) and one had some concerns ([Shah 2014](#)).

Net weight loss (kilograms)

[Risk of bias table for Analysis 1.1](#). Five trials were included in the analysis of net weight loss. Of these, one had low risk of bias ([Felker 2011](#)), two had some concerns ([Allen 2010](#); [Yayla 2015](#)), and two had high risk of bias ([Sager 2020](#); [Zheng 2021](#)).

All-cause mortality

[Risk of bias table for Analysis 1.2](#). Five trials were included in the analysis of all-cause mortality. Of these, one had low risk of bias ([Felker 2011](#)), two had some concerns ([Allen 2010](#); [Yayla 2015](#)), and two had high risk of bias ([Sager 2020](#); [Zheng 2021](#)).

Cardiovascular mortality

No studies included in the review reported this outcome.

Length of hospital stay (days)

[Risk of bias table for Analysis 1.3](#). Four studies were included in the analysis of length of hospital stay. Of these, three had some concerns ([Allen 2010](#); [Shah 2014](#); [Yayla 2015](#)), and one had high risk of bias ([Zheng 2021](#)).

Readmission to hospital following discharge

[Risk of bias table for Analysis 1.4](#). Three studies were included in the analysis of readmission to hospital. One had low risk of bias ([Felker 2011](#)), one had some concerns ([Shah 2014](#)), and one had high risk of bias ([Sager 2020](#)).

Occurrence of acute kidney injury

[Risk of bias table for Analysis 1.5](#). Three studies were included in the analysis of occurrence of acute kidney injury. Of these, two had low risk of bias ([Felker 2011](#); [Llorens 2014](#)), and one had high risk of bias ([Zheng 2021](#)).

Other potential sources of bias

No relevant other potential bias was identified.

Effects of interventions

See: [Summary of findings 1 Summary of findings table - Continuous infusion of intravenous loop diuretics compared to bolus injection of intravenous loop diuretics for adults \(aged greater than 18 years\) with acute heart failure](#)

See [Summary of findings 1](#).

1. Primary outcomes

1.1 Daily fluid balance (millilitres)

One study reported the daily fluid balance (daily fluid input volume minus urinary output volume) over 96 hours ([Shah 2014](#); 60 participants over two arms included in our analysis). From the start of intravenous diuretic treatment, they found 'significantly' higher daily fluid balance, in keeping with reduced diuresis, within the first 24 hours in the continuous infusion group compared to the bolus injection group (MD 395.58 mL, 95% CI 90.10 to 701.06; $P = 0.01$). Whereas there was no 'significant' difference in fluid balance beyond this time point (24 to 48 hours: MD -235.74 mL, 95% CI -671.85 to 200.37; $P = 0.29$; 48 to 72 hours: MD -194.03, 95% CI -507.86 to 119.80; $P = 0.23$; 72 to 96 hours: MD -33.96 mL, 95% CI -166.52 to 98.60; $P = 0.62$).

[Llorens 2014](#) (109 participants in total) reported the fluid balance at the first 24 hours, as well as at 3 hours, 6 hours, and 12 hours from the start of diuretics. They reported a 'significant' difference among three treatment groups (group 1: continuous infusion of furosemide at 10 mg/hour; group 2: furosemide bolus injection at 20 mg every 6 hours; group 3: furosemide bolus injection at 20 mg every 8 hours) in the net fluid balance over 24 hours. However, this outcome was solely presented in a figure, with no further data accessible attempts to contact the study author.

Meta-analysis of this outcome was therefore not performed. The evidence is very uncertain about the effect of continuous infusion

of intravenous loop diuretics compared to intravenous injection of loop diuretics on daily fluid balance in millilitres.

Additionally, two studies reported accumulative net urinary output over 72 hours (accumulative urinary output volume minus fluid intake volume) (Felker 2011; Zheng 2021; see Table 1). The difference in accumulative net urinary output between the continuous infusion and bolus injection of furosemide was observed as 'significant' by one study, favouring continuous infusion (Zheng 2021; 81 participants; $P = 0.01$), but 'insignificant' in the other (Felker 2011; 308 participants; $P = 0.89$).

1.2 Net weight loss (kilograms)

As rationalised in the Differences between protocol and review section, we pooled the net weight loss for five studies (Allen 2010; Felker 2011; Sager 2020; Yayla 2015; Zheng 2021). Continuous intravenous infusion of loop diuretics may result in mean net weight loss (from admission to discharge) of 0.86 kg more than bolus injection of intravenous loop diuretics, but the evidence is very uncertain (MD 0.86 kg, 95% CI 0.44 to 1.28; 5 trials, 497 participants; $P < 0.001$, $I^2 = 21\%$; very low-certainty evidence; Analysis 1.1).

Sensitivity analysis excluding two studies with high risk of selection and attrition bias (Sager 2020; Zheng 2021), showed insufficient evidence for a difference in net weight loss between participants who received continuous infusion and those who received bolus injection of loop diuretics (MD 0.70 kg, 95% CI -0.06 to 1.46; 3 trials, 378 participants; $P = 0.07$, $I^2 = 0\%$; Analysis 2.1).

1.3 All-cause mortality

Five studies reported all-cause mortality (Allen 2010; Felker 2011; Sager 2020; Shah 2014; Zheng 2021). There may be little to no difference in all-cause mortality between continuous infusion and bolus injection (RR 1.53, 95% CI 0.81 to 2.90; 5 trials, 530 participants; $P = 0.19$, $I^2 = 4\%$; low-certainty evidence; Analysis 1.2).

Sensitivity analysis excluding two studies with high risk of selection and attrition bias (Sager 2020; Zheng 2021, respectively), showed insufficient evidence for a difference in all-cause mortality between participants who received continuous infusion and those who received bolus injection of loop diuretics (RR 1.22, 95% CI 0.64 to 2.35; 3 trials, 409 participants; $P = 0.55$, $I^2 = 0\%$; Analysis 2.2).

1.4 Cardiovascular mortality

No studies reported cardiovascular mortality.

2. Secondary outcomes

2.1 Length of hospital stay (days)

Four studies reported length of hospital stay (Allen 2010; Shah 2014; Yayla 2015; Zheng 2021). There may be little to no difference in the length of hospital stay between participants who received continuous infusion and those who received bolus injection of loop diuretics, but the evidence is very uncertain (MD -1.10 days, 95% CI -4.84 to 2.64; 4 trials, 211 participants; $P = 0.57$, $I^2 = 88\%$; very low-certainty evidence; Analysis 1.3). There was significant heterogeneity noted among the studies; however, we were unable to perform subgroup analysis for this outcome due to lack of sufficient information from studies in the prespecified parameters, such as the percentage of participants with CKD, and

the percentage of a new diagnosis of AHF among participants. Therefore, we could not identify where the heterogeneity possibly derived from.

Sensitivity analysis excluding one study with high risk of attrition bias (Zheng 2021) noted improved heterogeneity among the studies; however, there was still insufficient evidence for a difference in the length of hospital stay between participants who received continuous infusion and those who received bolus injection of loop diuretics (MD 0.61 days, 95% CI -1.53 to 2.75; 3 trials, 130 participants; $P = 0.58$, $I^2 = 42\%$; Analysis 2.3).

2.2 Readmission to hospital following discharge for the longest available follow-up stated in the trials

Three studies reported readmission to hospital following discharge for the longest available follow-up (Felker 2011; Sager 2020; Shah 2014). There may be little to no difference in the readmission to hospital between participants who received continuous infusion and those who received bolus injection of loop diuretics (RR 0.85, 95% CI 0.63 to 1.16; 3 trials, 400 participants; $P = 0.31$, $I^2 = 0\%$; low-certainty evidence; Analysis 1.4).

Sensitivity analysis excluding one study with high risk of randomisation bias (Sager 2020) showed insufficient evidence for a difference in the readmission to hospital between participants who received continuous infusion and those who received bolus injection (RR 0.86, 95% CI 0.63 to 1.19; 2 trials, 368 participants; $P = 0.37$, $I^2 = 0\%$; Analysis 2.4).

2.3 Percentage change of B-type natriuretic peptide

No study reported the percentage change of BNP level among the participants.

One study found insufficient evidence of a difference in the change of BNP between participants who received continuous infusion of loop diuretics and those who received bolus injection (81 participants; mean -536 (standard error of the mean (SEM) 167.92) pg/mL with continuous infusion versus -488.35 (SEM 190.74) pg/mL with bolus; $P = 0.08$) (Zheng 2021).

2.4 Percentage change of N-terminal-proBNP

No study reported the percentage change of NT-proBNP level among the participants.

Two studies reported the change of NT-proBNP levels at 72 hours of treatment (Felker 2011) or on discharge (Sager 2020). Among them, the only study that analysed the outcome between the treatment groups found insufficient evidence of a difference in the change of the NT-proBNP levels between participants who received continuous infusion and those who received bolus injection of loop diuretics (308 participants; -1773 (SD 3828) pg/mL with continuous infusion versus -1316 (SD 4364) pg/mL with bolus; $P = 0.44$) (Felker 2011).

2.5 Adverse events (incidence of ototoxicity, electrolyte imbalances, hypotension, acute kidney injury)

2.5.1 Ototoxicity

One study reported tinnitus as an endpoint (Zheng 2021). They found insufficient evidence of a difference of ototoxicity incidence in participants receiving continuous infusion of intravenous loop diuretics compared with bolus injection of loop diuretics (1/42

(2.4%) participants with continuous infusion versus 3/39 (7.7%) participants with bolus; $P = 0.56$).

2.5.2 Electrolyte imbalances

Allen 2010 reported insufficient evidence of a difference in the incidence of hypokalaemia (less than 3.5 mEq/L) with bolus injection compared to continuous infusion group (2/20 (10%) participants with continuous infusion versus 5/21 (24%) participants with bolus; $P = 0.44$).

Zheng 2021 reported on the incidence of hyponatraemia, hypokalaemia, and hypomagnesaemia. They reported insufficient evidence of a difference in the incidence of all three (hyponatraemia: 8/42 (19.1%) participants with continuous infusion versus 5/39 (12.8%) participants with bolus; $P = 0.45$; hypokalaemia: 10/42 (23.8%) participants with continuous infusion versus 7/39 (17.9%) participants with bolus; $P = 0.52$; hypomagnesaemia: 12/42 (28.6%) participants with continuous infusion versus 9/39 (23.8%) participants with bolus; $P = 0.57$).

Felker 2011 reported combined incidence of serious adverse events including metabolic serious adverse events in the form of gout, hyperkalaemia, hypokalaemia, and hyponatraemia. There was no specific analysis reported on the outcomes of electrolyte imbalance that was relevant to our review. However, the trial reported the incidence of hyperkalaemia as 6/152 (4%) with continuous infusion versus 2/156 (1%) with bolus, incidence of hypokalaemia as 1/152 (1%) with continuous infusion versus 1/156 (1%) with bolus, and incidence of hyponatraemia as 1/152 (1%) with continuous infusion versus 1/156 (1%) with bolus.

2.5.3 Hypotension

One study reported on hypotension as an endpoint but failed to provide the clinical definition for hypotension adopted in the trial (Zheng 2021). They reported insufficient evidence of a difference in the incidence of hypotension between the continuous infusion group and the bolus injection group (5/42 (11.9%) participants with continuous infusion versus 3/39 (7.7%) participants with bolus; $P = 0.53$).

2.5.4 Acute kidney injury

Three trials included acute kidney injury as an adverse event outcome measure (Felker 2011; Llorens 2014; Zheng 2021). Felker 2011 and Llorens 2014 defined acute kidney injury as a rise in creatinine greater than 0.3 mg/dL, whereas Zheng 2021 did not provide a definition. There may be little to no difference between continuous infusion of intravenous loop diuretics and bolus injection of intravenous loop diuretics in the occurrence of acute kidney injury as an adverse event (RR 1.02, 95% CI 0.70 to 1.49; 3 trials, 491 participants; $P = 0.92$, $I^2 = 0\%$; low-certainty evidence; Analysis 1.5).

Sensitivity analysis excluding one study with high risk of bias due to missing outcome data (Zheng 2021) still showed that continuous infusion of intravenous loop diuretics may not reduce the occurrence of acute kidney injury as an adverse event compared to the bolus injection of intravenous loop diuretics (RR 1.10, 95% CI 0.72 to 1.68; 2 trials, 410 participants; $P = 0.65$, $I^2 = 0\%$; Analysis 2.5).

Additionally, Allen 2010 reported no difference in creatinine change (41 participants; 0.13 (SD 0.34) mg/dL with continuous

infusion versus -0.02 (SD 0.39) mg/dL with bolus; $P = 0.18$). Sager 2020 analysed the change of creatinine level from enrolment to discharge within the same group, and reported insufficient evidence of a difference in the change of mean creatinine level in either group during hospitalisation (continuous infusion: -7.75 (standard error (SE) 8.3) μ mol/L, 20 participants, $P = 0.36$; bolus injection group -0.1 (SE 8.3) μ mol/L, 20 participants, $P = 0.99$). Yayla 2015 described a prespecified study endpoint of changes in serum creatinine level from baseline to 48 hours. In comparing their three groups using an analysis of variance (ANOVA) one-way test, they found no difference in the mean change in creatinine level (continuous infusion; 0.16 (SD 0.21) mg/dL, bolus; 0.04 (SD 0.15) mg/dL, hypertonic saline group; 0.20 (SD 0.21) mg/dL; $P = 0.08$). However, they described a statistically 'significant' ($P = 0.03$) post-hoc analysis using a Tukey's test comparing the bolus and infusion subgroups separately.

2.6 Change in six-minute walk test (metres)

No study reported a six-minute walk test.

2.7 Change in visual analogue scale for dyspnoea (10-cm scale)

One study used a visual analogue scale of dyspnoea as a measure of participants' global assessment of symptoms (Felker 2011). Participants were asked to evaluate their general well-being by marking a 10-cm vertical line, with the top labelled "best you have ever felt" and the bottom labelled "worst you have ever felt". They scored the participants' markings on a scale of 0 to 100 by measuring the distance in millimetres from the bottom of the line. Participants self-assessed dyspnoea at randomisation, and 6, 12, 24, 48, 72, and 96 hours. The results were quantified as the area under the curve (AUC) of the score on the visual analogue scale.

They found no 'significant' difference in their primary efficacy endpoint of participants' global assessment of symptoms with bolus injection of intravenous loop diuretics compared to continuous infusion of loop diuretics (308 participants; mean AUC: 4373 (SD 1404) with continuous infusion versus 4236 (SD 1440) with bolus; $P = 0.47$).

DISCUSSION

In this review, we analysed seven RCTs including 681 participants addressing a comprehensive list of outcomes.

We performed meta-analysis on five outcomes (net weight loss: Analysis 1.1; all-cause mortality: Analysis 1.2; length of stay in hospital: Analysis 1.3; readmission to hospital following discharge: Analysis 1.4; occurrence of acute kidney injury as an adverse event: Analysis 1.5). We performed sensitivity analysis on the same outcomes but with the exclusion of trials with a high risk of bias. See Summary of findings 1 and Summary of main results for more details.

The topic of continuous versus bolus injection of intravenous diuretics has been widely debated in the literature for many years, but clear evidence of superiority of either delivery method has not yet been established. The safety and efficacy profile of these two methods are similar and each have their own unique advantages.

From a pharmacological perspective, continuous intravenous infusions of diuretics offer potential advantages over bolus dosing. It is postulated that lower peak plasma concentrations of diuretics

with continuous infusions may lead to a reduced incidence of adverse effects, electrolyte imbalances, and a more stable clinical haemodynamic profile. Furthermore, the constant delivery of medication may prevent rebound of sodium reabsorption that can occur when peak plasma concentrations of diuretics decline after a bolus injection. In contrast, bolus injections allow for greater patient mobility and the possibility of treatment in non-hospital environments such as community ambulatory units or in patients' own homes.

After conducting an exhaustive analysis of existing data comparing the continuous infusion of loop diuretics with bolus intermittent administration in people with AHF, we found that the current data are insufficient to show superiority of either delivery method. To ensure the validity of our findings, we only included trials that met our strict inclusion and exclusion criteria. While previous studies have suggested the benefits of continuous infusion over bolus injections, we believe that our review, which excluded trials with clinical confounders and RCTs with a high risk of bias, provides the most robust conclusion to date.

Summary of main results

This review examined the effects of intravenous continuous infusion versus bolus injection of loop diuretics for the initial treatment of AHF in an adult hospitalised population. We included data from seven RCTs and scrutinised the data for risk of bias and assessed the certainty of evidence using rigorous, recognised Cochrane review methods.

The *main* outcomes investigated by the studies included net weight loss, all-cause mortality, length of hospital stay, readmission to hospital following discharge, and occurrence of acute kidney injury. To further validate our findings, and make them relevant to daily clinical practice, we excluded trials with high risk of bias from our sensitivity analyses and descriptions below.

Net weight loss

Following exclusion of two studies with high risk of selection and attrition bias, we included three trials with 378 participants in this analysis (Allen 2010; Felker 2011; Yayla 2015). Two of the remaining trials did retain some concerns for risk of bias in randomisation (Allen 2010) and in selection of the reported result (Allen 2010; Yayla 2015). Sensitivity analysis revealed insufficient evidence for a difference in net weight loss between participants who received continuous infusion and those who received bolus injection of loop diuretics.

All-cause mortality

Following exclusion of two studies with high risk of selection and attrition bias, we included three trials with 409 participants in this analysis (Allen 2010; Felker 2011; Shah 2014). Two of the remaining trials did retain some concerns for risk of bias in randomisation due to deviations from intended interventions (Shah 2014) and in selection of the reported result (Allen 2010; Shah 2014). Sensitivity analysis revealed insufficient evidence for a difference in all-cause mortality between participants who received continuous infusion and those who received bolus injection of loop diuretics.

Length of hospital stay (days)

Following exclusion of one trial with high risk of attrition bias, we included three trials with 130 participants in this analysis (Allen

2010; Shah 2014; Yayla 2015). The remaining trials did retain some concerns for risk of bias in randomisation (Allen 2010; Shah 2014) and in selection of the reported result (Allen 2010; Shah 2014; Yayla 2015). Sensitivity analysis reveals insufficient evidence for a difference in the length of hospital stay between participants who received continuous infusion and those who received bolus injection of loop diuretics.

Readmission to hospital following discharge for the longest available follow-up stated in the trials

Following exclusion of one trial with high risk of randomisation bias, we included two trials with 368 participants in this analysis (Felker 2011; Shah 2014). One trial retained some concerns for risk of bias in randomisation, deviations from the intended interventions, and in selection of the reported result (Shah 2014). Sensitivity analysis showed insufficient evidence for a difference in the readmission to hospital between participants who received continuous infusion and those who received bolus injection.

Acute kidney injury

Following exclusion of one trial with high risk of bias due to missing outcomes data, we included two trials with 410 participants in this analysis (Felker 2011; Llorens 2014). Both trials were at low risk of bias and sensitivity analysis showed insufficient evidence for a difference in the occurrence of acute kidney injury as an adverse event between participants who received continuous infusion and those who received bolus injection.

Thus, taking into consideration the risk of bias and certainty of evidence, the evidence included in our review suggests that the use of continuous infusion over bolus injection of loop diuretics may result in little to no difference in outcomes across all outcomes. See [Summary of findings 1](#).

Overall completeness and applicability of evidence

Including RCTs with low- to very low-certainty evidence partially limits the application of our findings to clinical decision-making. We can observe this in our selective sensitivity analysis, whereby excluding studies with less than high-certainty evidence we observe different results. The available evidence comprehensively addresses the comparisons between continuous infusion and bolus injection of intravenous loop diuretics in hospitalised adults (aged greater than 18 years) and our intended PICO (Population, Intervention, Comparison, and Outcome). However, it is important to note that the applicability of this evidence to real-world clinical settings may vary due to factors such as study design limitations and the imprecision of the results. Nonetheless, this review provides the highest level of current evidence for the aim of this review.

Quality of the evidence

We thoroughly assessed each included study for each individual sensitivity analysis using the Cochrane RoB 2 tool (see [Risk of bias table for Analysis 1.1](#); [Risk of bias table for Analysis 1.2](#); [Risk of bias table for Analysis 1.3](#); [Risk of bias table for Analysis 1.4](#); [Risk of bias table for Analysis 1.5](#)). The overall risk of bias ranged from high to low and the overall certainty of evidence ranged from low to very low. To ensure that the outcomes deduced from this analysis are unbiased, we repeated sensitivity analysis and excluded trials with overall high risk of bias.

We also assessed the certainty of the evidence using GRADEpro GDT ([Summary of findings 1](#)). None of the prespecified outcomes had high- or moderate-certainty evidence. All-cause mortality, readmission to hospital, and occurrence of acute kidney injury outcomes had low-certainty evidence and the remaining outcomes had very low-certainty evidence. As discussed in [Effects of interventions](#), we only performed narrative analysis for 'daily fluid balance in millilitres', despite this, we downgraded evidence three levels due to 'study limitations, inconsistency and imprecision'. No studies measured cardiovascular mortality. We did not downgrade any studies for indirectness, and publication bias was undetected across all trials. The effects of plausible residual confounders, the magnitude of large-effect, and dose-response gradient were not relevant.

Inconsistency: as described above, we downgraded the narrative analysis for inconsistency due to one trial presenting a figure with no corresponding data. In addition, we downgraded the certainty of evidence for 'length of hospital stay' due to substantial unexplained heterogeneity.

Imprecision: for all outcomes, we downgraded the certainty of the evidence once for imprecision as pooled estimates were based on small sample sizes/low event rates and had wide 95% CIs that included the possibility of no effect.

Risk of bias: the certainty of evidence for:

- 'daily fluid balance in millilitres': downgraded one level due to risk of bias as one trial had some concerns for overall risk of bias';
- 'net weight loss': downgraded two levels due to risk of bias. There was high overall risk of bias in two trials and some concerns overall in another two trials included in the analysis;
- 'all-cause mortality': downgraded only one level due to risk of bias as one trial with high risk of bias did not have any events and the proportion of information from another trial with high risk of bias was insufficient to downgrade the certainty of evidence further;
- 'length of hospital stay': downgraded two levels due to risk of bias. There was overall high overall risk of bias in one trial and some concerns overall in the other three included trials;
- 'readmission to hospital following discharge': downgraded only one level due to risk of bias as the proportion of information obtained from one trial with high risk of bias was not sufficient to downgrade the certainty of evidence further;
- 'occurrence of acute kidney injury': downgraded only one level due to risk of bias as the proportion of information obtained from one trial with high risk of bias was not sufficient to downgrade the certainty of evidence further.

Potential biases in the review process

This review was conducted in keeping with the Cochrane Heart Group methodology and the protocol ([Zhang 2021](#)), which was prepublished in August 2021. However, partially due to the COVID pandemic, there was a significant delay in the completion and the revision of the review. In order to reduce the publication bias and to ensure an updated review on most recent evidence, we conducted a top-up search, which subsequently did not identify new studies to be included in the review. Cochrane Information Specialists developed the search strategy for both the initial search and the top-up search. We did not exclude any cross-over studies

from the bibliographical database searches, but manually excluded them during the initial screening and full-text assessment process in order to minimise any selection bias and publication bias. We did not apply any restrictions on the language of the publication during the literature search. We also attempted to minimise the effects of publication bias by conducting extensive searches for published and unpublished RCTs, reviewing bibliographies for additional RCTs, and by seeking missing data and clarification directly from study authors. However, due to low response rates from the corresponding authors, there was limited additional information obtained via correspondence for included studies, as well as studies awaiting classification. The studies that were still awaiting classification could have produced negative results, which might have affected their publication and their availability to the public, hence there is potential reporting bias remaining despite our best attempts.

We minimised selection bias by involving two review authors in study selection and data extraction, with arbitration by a third review author. A fourth review author performed independent study selection review and data validation for all relevant RCTs included in our review.

Most studies included in the review had predefined primary and secondary outcomes, and a large number of predefined outcomes could potentially cause a multiple comparison bias. We attempted to minimise this bias by conducting sensitivity analyses in keeping with the protocol ([Zhang 2021](#)).

None of the review authors were involved in the conduct of the included trials.

Agreements and disagreements with other studies or reviews

The previous Cochrane review had a different meta-analysis protocol and inclusion criteria that included cross-over RCTs ([Salvador 2005](#)). This research methodology did not form part of our analysis as we only included parallel RCTs. [Salvador 2005](#) concluded that greater diuresis could be obtained when intravenous loop diuretics were given as continuous infusion and that they displayed a better safety profile. These results seem entirely driven by inclusion of one trial that was confounded by the parallel use of hypertonic saline in the continuous infusion treatment arm (where the infusion only lasted 30 minutes, i.e. not an infusion by our definitions). Importantly, similar outcomes were not observable when this trial was excluded in a repeat sensitivity analysis.

While several other meta-analyses have indicated potential benefits of continuous infusion over bolus injections of diuretics, including enhanced weight reduction ([Alqahtani 2014](#); [Amer 2012](#); [Chan 2020](#); [Karedath 2023](#); [Kuriyama 2019](#); [Liu 2021](#)), improved early or daily diuresis measured by urine output ([Alqahtani 2014](#); [Amer 2012](#); [Chan 2020](#); [Kuriyama 2019](#); [Liu 2021](#); [Ng 2018](#)), reduced all-cause mortality ([Liu 2021](#)), and reduced heart-failure readmissions ([Liu 2021](#)), it is pertinent to note that these reviews included trials that did not align with the strict inclusion criteria of our review. For example, including cross-over trials, studies with confounding use of inotropes/vasopressors and hypertonic saline, and trials where hourly urine output served as a protocol-driven endpoint. To uphold the rigour of our review methodology further, we also deliberately

excluded trials with a high risk of bias in our sensitivity analysis. This precise approach guarantees a thorough examination and comparison of intravenous diuretic administration methods, free from confounding variables originating from divergent clinical practices and confounding use of additional therapies. Hence, our review stands as the most untainted evaluation of continuous infusion versus bolus injections of furosemide in adults in hospital.

AUTHORS' CONCLUSIONS

Implications for practice

After conducting a thorough analysis of existing data comparing the continuous infusion of loop diuretics with bolus intermittent administration in people with acute heart failure, we have found that the current evidence is insufficient to show superiority of one intervention over the other. To ensure the validity of our findings, we only included trials that met our strict inclusion and exclusion criteria.

While previous studies have suggested the benefits of continuous infusion over bolus injections, our review could not provide sufficient evidence to support or refute this, and we believe that our review, which excludes trials with clinical confounders and randomised controlled trials with a high risk of bias, provides the most robust and accurate conclusion to date.

Given the variability in medical practice across different regions of the world, healthcare providers may adopt approaches that best suits their experience and available infrastructure, staffing, experience, finances, and equipment.

Implications for research

We propose that future research in the field should shift its focus away from delivery methods and instead concentrate on the usability of these different methods outside the traditional hospital setting, as the evidence suggests that similar outcomes can be achieved with either method in a select patient population. As the described hard outcomes are neutral, it is crucial to also consider other important markers such as patient-reported quality-of-life measures and patient preference, which have been unexplored in the literature to date. Indeed, these markers may become deciding factors in the future, as they promote patient-centred and holistic care.

Moreover, treating people in an ambulatory setting or via a virtual ward at home may lead to a reduction in potential

adverse effects of hospitalisation such as prolonged wait in the emergency department, hospital-acquired infections, healthcare-related delirium, falls, reduced nutrition, and muscle deconditioning. With the aid of digital technology, use of lung ultrasound to assess congestion, and point-of-care testing including finger-prick electrolyte monitoring and urinary spot-sodium testing, patients can receive close monitoring of their diuresis management in these settings. These methods, coupled with an 'early-supported discharge' approach may result in financial benefits by reducing healthcare costs and preserving hospital beds for critically ill patients. Even if these measures prove to be cost-neutral in the future, the impact on patients' quality of life could become the decisive factor in decision-making.

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Editorial contributions

Cochrane Heart Group supported the authors in the development of this systematic review. The following people conducted the editorial process for this article.

Sign-off Editor (final editorial decision): Prof Michael Brown, Michigan State University College of Human Medicine, USA; **Managing Editor** (selected peer reviewers, collated peer-reviewer comments, provided editorial guidance to authors, edited the article): Marwah Anas El-Wegoud, Cochrane Central Editorial Service; **Editorial Assistant** (conducted editorial policy checks and supported editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service; **Editorial Assistant** (identified peer reviewers): Kylie Bade, Central Editorial Services; **Copy Editor** (copy editing and production): Anne Lawson, Cochrane Central Production Service.

Peer-reviewers (provided comments and recommended an editorial decision): Bradley A Bart, MD Minneapolis VAHCS and the University of Minnesota (**clinical review**); Philippe Meyer, MD, Cardiology Division, University Hospital of Geneva, Switzerland (**clinical review**); Nuala Livingstone, Cochrane Evidence Production and Methods Directorate (**methods review**); Jo Platt, Central Editorial Information Specialist (**search review**); Brian Duncan (**consumer review**).

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CHARACTERISTICS OF STUDIES

Characteristics of included studies [ordered by study ID]

Allen 2010

Study characteristics

Methods	Randomised controlled trial
	Participants enrolled from Duke University Medical Center from 1999 to 2005
	Treatment location: hospital
	Country: USA
	Duration of study: 6 years
Participants	41 in total, 21 received bolus, 20 received continuous infusion
	Mean age: 60 years
	Women (%): 37%

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Allen 2010 (Continued)

	Lost to follow-up: 0 Diagnostic criteria for ADHF: clinical diagnosis with evidence of volume overload (pulmonary congestion on chest x-ray or pro B-type natriuretic peptide greater than the upper limit of normal for age) Subtypes of ADHF: not specified Mean LVEF: 35%
Interventions	Continuous infusion: intravenous furosemide dosage decided by attending physician (mean 162 mg/24 hours (SD 52 mg/24 hours)), administered as continuous infusion. Bolus injection: intravenous furosemide dosage decided by attending physician (mean 162 mg/24 hours (SD 48 mg/24 hours)), administered as bolus injections twice daily. Treatment duration: fixed dosage for the initial 48 hours. Subsequent titration of furosemide dose was at the discretion of the attending physician but was guided by a dose-escalation algorithm.
Outcomes	Primary endpoint: change in creatinine from admission to hospital day 3 or discharge Secondary endpoint: cumulative urine output, electrolyte changes from admission to day 3 on admission or discharge, length of hospitalisation, death within 90 days
Notes	Trial registry number: not applicable Funding source: none declared Conflict and interest: none declared

Felker 2011

Study characteristics

Methods	Multicentre double-blind double-dummy randomised controlled trial Study adopted 2 × 2 factorial design. Participants were randomly assigned, in a 1:1:1:1 ratio, to low-dose strategy (total intravenous furosemide dose equal to their total daily oral loop diuretic dose in furosemide equivalents) or a high-dose strategy (total daily intravenous furosemide dose 2.5 times their total daily oral loop diuretic dose in furosemide equivalents) and to administration of furosemide either by intravenous bolus every 12 hours or by continuous intravenous infusion. Country: 26 centres in USA and Canada Start date: March 2008 End date: November 2009 Duration of the study: 20 months
Participants	Adults with signs and symptoms consistent with ADHF within 24 hours of hospital admission Total = 308 participants, 156 received bolus injections, 152 received continuous infusion (low-dose diuretics = 151, high-dose diuretics = 157) Mean age: 66 years Women (%): 27% Lost to follow-up: 0

Felker 2011 (Continued)

Diagnostic criteria for ADHF: the presence of ≥ 1 symptom (dyspnoea, orthopnoea, or oedema) and ≥ 1 sign (rales, peripheral oedema, ascites, or pulmonary vascular congestion on chest radiography) of heart failure

Subtypes of ADHF: deterioration of chronic heart failure

Mean LVEF: 35%

Interventions	Continuous infusion: - high-dose group: continuous intravenous infusion of furosemide at the dose 2.5 times the total daily oral furosemide; - low-dose group received total intravenous furosemide dose equal to their total daily oral loop diuretic dose in furosemide equivalents. Bolus injection: - high-dose group: total dosage of furosemide equivalent to 2.5 times the total daily oral furosemide to be administered in 2 boluses every 12 hours; - low-dose group: intravenous furosemide equivalent to the total daily oral loop diuretic dose administered in 2 intravenous boluses at 12 hours interval. Treatment duration: 72 hours Treatment location: emergency department (24 hours) then on wards
Outcomes	Primary efficacy endpoint: participants' global assessment of symptoms (visual analogue scale), quantified as area under the curve of serial assessments from baseline to 72 hours. Primary safety endpoint: change in the serum creatinine level from baseline to 72 hours. Secondary endpoints: participant-reported dyspnoea, change in bodyweight, net fluid loss, proportion of participants who were free from congestion at 72 hours, worsening renal function (defined as increase in creatinine level by 0.3 mg/dL), worsening or persistent heart failure, treatment failure, change in biomarker levels at 72 hours, day 7 or discharge, and day 60. Clinical endpoint as composite of death, rehospitalisation, or an emergency department visit within 60 days, composite of total number of days hospitalised or dead within 60 days after randomisation.
Notes	Trial registry number: NCT00577135 Funding source: National Heart, Lung, and Blood Institute Conflict and interest: - Dr Felker reports receiving consulting fees from Amgen, Cytokinetics, Corthera, Otsuka, Novartis, and Roche Diagnostics and grant support from Amgen, Cytokinetics, Otsuka, and Roche Diagnostics; - Dr LeWinter, receiving consulting fees from Novartis, grant support from Medtronic, and lecture fees from Medtronic and Novartis; - Dr Anstrom, receiving consulting fees from Johnson & Johnson and Pfizer; - Dr Hernandez, receiving consulting fees from Amgen and grant support from Johnson & Johnson and serving on a clinical endpoints committee for Corthera; - Dr Velazquez, receiving consulting fees from Novartis and Boehringer Ingelheim, grant support from Johnson & Johnson, and lecture fees from Novartis; - Dr Kfoury, receiving grant support from Novartis and XDx; - Dr Chen, receiving grant support from Scios, Anexon, and Nile Therapeutics, being named as a coinventor on patents for chimeric natriuretic peptides, and receiving royalties from Nile Therapeutics; - Dr O'Connor, receiving consulting fees from Merck, GE Healthcare, Forest Pharmaceuticals, Medtronic, Novella Clinical, Medpace, Roche, Actelion Pharmaceuticals, Amgen, Trevena, and Martek Biosciences and grant support from Johnson & Johnson.

Felker 2011 (Continued)

No other potential conflict of interest relevant to this article was reported.

Llorens 2014
Study characteristics

Methods	Randomised controlled trial Country: Spain Start date: January 2009 End date: December 2009 Duration of the study: 12 months
Participants	Acute decompensated heart failure admitted to emergency department 109 in total, 36 received continuous infusion, 37 received bolus every 6 hours, and 36 received bolus injection every 8 hours Mean age: 82 years Women (%): 66% Lost to follow-up: 0 Diagnostic criteria for ADHF: according to European Society of Cardiology guideline: signs, symptoms, structural or functional signs of myocardial dysfunction Subtypes of ADHF: not specified Mean LVEF: not reported; however, 50% participants have reduced LVEF.
Interventions	Continuous infusion: furosemide initial bolus of 40 mg before randomisation, followed by furosemide 10 mg/hour given by continuous intravenous infusion over 24 hours Bolus injection: furosemide intravenous boluses of 20 mg every 6 or 8 hours in 2 separate groups Treatment duration: 24 hours Treatment location: observation unit of emergency department
Outcomes	Primary endpoint: urinary output and net diuresis (urine output minus intravenous volume) at 3, 6, 12, and 24 hours Secondary endpoint: improvement in dyspnoea, presence of orthopnoea, improvement in peripheral oedema, improvement in the extension of rales at pulmonary auscultation, systolic and diastolic blood pressure, heart rate, respiratory rate, oxygen saturation at 3, 6, 12, and 24 hours Safety endpoint: increase in creatinine of > 0.3 mg/dL, development of hyponatraemia or hypokalaemia
Notes	Trial registry number: EudraCT 2008-004488-20 Funding source: none declared Conflict and interest: none declared

Sager 2020

Study characteristics

Methods	Randomised controlled trial Country: Sweden Start date: January 2018 End date: September 2018 Duration of the study: 8 months
Participants	Adults hospitalised to the Internal Medicine ward with worsened symptoms of their heart failure 40 participants in total, 20 received continuous infusion, 20 received bolus injections Mean age: 82 years Women (%): 60% Lost to follow-up: 0 Diagnostic criteria for ADHF: not specified Subtypes of ADHF: not specified Mean LVEF: not reported
Interventions	Continuous infusion: furosemide 100–500 mg dissolved in 100–250 mL of sodium chloride 9% solution (as reported in the publication) to be administered continuously as infusions over 4–10 hours a day Bolus injection: given 1 or several times a day in total daily doses of furosemide 20–100 mg Treatment duration: not specified Treatment location: internal medicine ward
Outcomes	Mortality at 3 months following discharge; NT-pro BNP, weight, creatinine, eGFR at enrolment and discharge; readmission after 30 days
Notes	Trial registry number: not applicable Funding source: Swedish Heart-Lung Foundation and the Swedish SUS Funds Conflict and interest: none declared

Shah 2014

Study characteristics

Methods	Randomised controlled trial Country: India Start date: April 2010 End date: June 2012
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Shah 2014 (Continued)

	Duration of the study: 26 months
Participants	Adults admitted to emergency department with ADHF 90 in total (60 included in the review), 30 received bolus injections, 30 received continuous infusion (excluded 30 participants who received continuous infusion plus dopamine) Mean age: 59 years* Women (%): 23%* Lost to follow-up: 0 Diagnostic criteria for ADHF: defined by ≥ 1 symptom (dyspnoea, orthopnoea, or oedema) and ≥ 1 sign (rales on auscultation, peripheral oedema, ascites) or pulmonary vascular congestion on chest radiography Subtypes of ADHF: not specified Mean LVEF: 33%
Interventions	Continuous infusion: initial furosemide 40 mg bolus in emergency department, followed with intravenous furosemide continuous infusion 100 mg/24 hours (intravenous furosemide 100 mg in 10 mL was dissolved in 14 mL of 0.9% normal saline to form a solution of 24 mL, given at the rate of 1 mL/hour infusion). Bolus injection: intravenous furosemide bolus 100 mg/24 hours in 2 divided doses. Treatment duration: not specified, but ≥ 48 hours beyond which the treating physician could adjust the diuretic strategy based on clinical responses. Treatment location: intensive cardiac care unit
Outcomes	Primary endpoint: negative fluid balance at 24 hours Secondary endpoint: duration of hospital stay; negative fluids balance at 48, 72, 92 hours; trend of serum sodium/potassium; creatinine and blood urea at 24, 48, 72 hours, 7 days, and 30 days; death; and emergency department visits in 30 days.
Notes	Trial registry number: not applicable Funding source: none declared Conflict and interest: none declared Described the study as "single-blind" trial; however, no relevant information identifiable from the publication. Email was sent to the corresponding author requesting more information.

Yayla 2015

Study characteristics

Methods	Randomised controlled trial Country: Turkey Start date: March 2011 End date: November 2012 Duration of the study: 20 months
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Yayla 2015 (Continued)

Participants	Adults hospitalised for ADHF with either reduced or preserved LVEF 43 in total (29 included in the review), 15 received continuous infusion, 14 received bolus injection (excluding 14 who received hypertonic saline alongside diuretics) Mean age: 68 years* Women (%): 48%* Lost to follow-up: 0 Diagnostic criteria for ADHF: not specified Subtypes of ADHF: not specified Mean LVEF: 43%*
Interventions	Continuous infusion: continuous infusion of furosemide 160 mg given over 16 hours/day. Bolus injection: intravenous bolus injections of furosemide 80 mg given twice daily. Treatment duration: 48 hours Treatment location: hospital
Outcomes	Change in weight; change in creatinine from baseline to 48 hours as well as from baseline to de-compensated state; length of hospitalisation; renal dysfunction defined as an increase in creatinine of > 0.3 mg/dL from baseline during treatment.
Notes	Trial registry number: not applicable Funding source: none declared Conflict and interest: none declared

Zheng 2021

Study characteristics

Methods	Randomised controlled trial Country: China Start date: June 2016 End date: July 2019 Duration of the study: 3 years
Participants	Adults with ADHF and moderate chronic renal insufficiency admitted to hospital 94 in total, 47 received continuous infusion, 47 received bolus injection Mean age: 57 years Women (%): 35% Lost to follow-up: 13 in total, 5 in continuous infusion, 8 in bolus injection Diagnostic criteria for ADHF: not specified Subtypes of ADHF: acute heart failure

Zheng 2021 (Continued)

Mean LVEF: 57%

Interventions	Continuous infusion: furosemide 160 mg for group A and 200 mg for group B diluted in 50 mL of 0.9% sodium chloride, and given over 6 hours as an infusion every day. Bolus injection: furosemide 160 mg for group A and 200 mg for group B diluted in 50 mL of 0.9% sodium chloride, and given within 5 minutes every day. Treatment duration: 72 hours Treatment location: nephrology ward
Outcomes	Primary endpoint: freedom from congestion at 72 hours (defined as jugular venous pressure < 8 cm without orthopnoea and presence of trace or no peripheral oedema) Secondary endpoint: net daily urine output; weight loss during the study; total urinary sodium excretion; length of hospital stay; adverse events including tinnitus, acute kidney injury, electrolyte disturbances during the treatment period
Notes	Trial registry number: not applicable Funding source: none declared Conflict and interest: none declared

*: participants included in the Cochrane review.

ADHF: acute decompensated heart failure; LVEF: left ventricular ejection fraction; SD: standard deviation.

Characteristics of excluded studies [ordered by study ID]

Study	Reason for exclusion
Aaser 1997	Ineligible study design: cross-over study.
Alvarez 2007	Ineligible population: people with fluid overload in intensive care unit, including but not exclusively acute decompensated heart failure.
Bagatin 1993	Ineligible study design: cross-over study.
	Ineligible intervention: intravenous infusion of loop diuretics over 1 hour.
ChiCTR1800017270	Ineligible intervention: hypertonic saline plus furosemide versus furosemide alone.
Dormans 1996	Ineligible study design: cross-over study.
Dormans 1997	Ineligible study design: cross-over study.
EUCTR2013-000866-12-IT	Ineligible intervention: inotropes used alongside intravenous loop diuretics.
Frea 2020	Ineligible intervention: inotropes used alongside intravenous loop diuretics.
Fudim 2021	Ineligible study design: retrospective analysis using participants from ASCEND-HF (Acute Study of Clinical Effectiveness of Nesiritide in Decompensated Heart Failure) trial with a stable diuretic strategy in the first 24 hours following randomisation.
Kelly 2015	Ineligible intervention.
Kramer 1996	Ineligible population: compensated stable heart failure.

Continuous infusion versus bolus injection of loop diuretics for acute heart failure (Review)

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Study	Reason for exclusion
Lahav 1992	Ineligible study design: cross-over design.
McFarland 1968	Ineligible intervention: oral furosemide.
Moon 2012	Ineligible study design: summary article for Felker 2011 (publication in <i>New England Journal of Medicine</i>).
NCT00904488	Ineligible intervention: oral metolazone and intermittent intravenous furosemide versus continuous infusion furosemide.
NCT01441245	Ineligible intervention: use of inotropes and hypertonic saline alongside furosemide.
NCT03592836	Ineligible intervention: inotropes used alongside intravenous loop diuretics.
NCT03892148	Ineligible intervention: use of loop diuretics and thiazide diuretics as per attending physician versus loop diuretics and thiazide diuretics as per CARRESS-HF (Cardiorenal Rescue Study in Acute De-compensated Heart Failure) protocol.
NCT03999216	Ineligible intervention: loop diuretics versus loop plus thiazide diuretics.
Palazzuoli 2013	Ineligible intervention: use of inotropes and hypertonic saline alongside furosemide.
Palazzuoli 2014a	Ineligible intervention: use of inotropes and hypertonic saline alongside furosemide.
Palazzuoli 2014b	Ineligible intervention: use of inotropes and hypertonic saline alongside furosemide.
Palazzuoli 2015a	Ineligible intervention: use of inotropes and hypertonic saline alongside furosemide.
Palazzuoli 2015b	Ineligible intervention: use of inotropes and hypertonic saline alongside furosemide.
Pivac 1998	Ineligible study design: cross-over study.
Ragab 2017	Ineligible intervention: inotropes used in both arms.
Ragab 2018	Ineligible intervention: inotropes used in both arms.
Ruocco 2019	Ineligible study design: retrospective analysis of low-dose loop diuretics versus high-dose diuretics.
Schuller 1997	Ineligible intervention type. They adjusted diuretic therapy by either continuous or bolus infusions of furosemide, titrated to achieve negative hourly fluid balance.
Shah 2012	Ineligible study design: post hoc analysis of the DOSE-AHF (Diuretic Optimization Strategies Evaluation in Acute Heart Failure) trial data, with a focus on the association of regular oral diuretic dosages with outcomes and response to intravenous diuretics.
Taema 2018	Ineligible intervention: vasopressor and inotropes used alongside loop diuretics.
Thomson 2010	Ineligible intervention: inotropes used alongside loop diuretics.
van Meyel 1993	Ineligible study design: cross-over study
Verel 1964	Ineligible comparison and intervention: comparison of oral furosemide of variable doses alongside other diuretics.

Characteristics of studies awaiting classification *[ordered by study ID]*

Cienki 2009

Methods	Randomised controlled trial
Participants	People with acute decompensated congestive heart failure with New York Heart Association Class II or above symptoms 20 participants randomised
Interventions	Continuous infusion vs bolus injection of furosemide. Dosage not clarified in the abstract. Quote: "Initial orders were written by the ED [emergency department] attending and method of administration of furosemide was maintained throughout the hospital." Treatment location: emergency department
Outcomes	Length of treatment
Notes	

Ho 2014

Methods	Randomised controlled trial
Participants	People admitted for acute decompensated heart failure < 24 hours with left ventricular ejection fraction $\geq 50\%$ 54 participants in total, 27 in each group Mean age: 79 years Women (%): 67%
Interventions	Quote: "Continuous infusion of frusemide (8mg/hr for 48hr) with intermittent bolus (80mg every 12hr for 4 doses)."
Outcomes	Primary endpoints were changes in B-type natriuretic peptide and neutrophil gelatinase-associated lipocalin in 48 hours. Secondary endpoints were: worsening of renal function defined by a rise in serum creatinine by $26.5 \mu\text{mol/L}$ or 50%, haemodynamic status assessed by serial quantitative echocardiography and body fluid status by bioimpedance vector analysis.
Notes	

Sharma 2018

Methods	Single-blind (investigators) randomised controlled trial Country: USA Start date: October 2013 End date: December 2016
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Sharma 2018 (Continued)

Duration of the study: 3 years

Participants	Adults with heart failure with preserved ejection fraction admitted with decompensated heart failure 90 in total (42 included in the review), 23 received continuous infusion, 19 received bolus injection (excluded from the review: 24 received bolus injection and dopamine, 24 received continuous infusion and dopamine) Mean age: 65 years Women (%): 69% Lost to follow-up: 0 Diagnostic criteria for acute decompensated heart failure: defined by the presence of ≥ 1 symptom (dyspnoea, orthopnoea, or oedema) AND 1 sign (rales on auscultation, elevated jugular venous pulse, hepatojugular reflex, peripheral oedema, ascites, pulmonary vascular congestion on chest radiography). Subtypes of acute decompensated heart failure: deterioration of chronic heart failure. Mean left ventricular ejection fraction: not reported; however, participants were reported as (quote) "have preserved ejection fractions on recruitment."
Interventions	Continuous infusion: intravenous furosemide with a recommended total daily dose equal to 2.0 times their total daily outpatient oral furosemide-equivalent dose; participants naive to outpatient loop diuretics receive a starting intravenous furosemide dose of 80 mg/day. Participants received total diuretic dose as a continuous infusion over 24 hours without a bolus dose. Bolus injection: intravenous furosemide with a recommended total daily dose equal to 2.0 times their total daily outpatient oral furosemide-equivalent dose; participants naive to outpatient loop diuretics receive a starting intravenous furosemide dose of 80 mg/day. Participants received the total daily diuretic dose divided into 2 doses every 12 hours. Treatment duration: ≥ 72 hours Treatment location: hospital
Outcomes	Primary endpoint: percentage change in creatinine at 72 hours Secondary endpoint: percentage change in cystatin, net volume, output volume, systolic blood pressure change, weight change, and worsening of renal function
Notes	Trial registry number: NCT01901809 Funding source: none declared Conflict and interest: none declared

Characteristics of ongoing studies [ordered by study ID]

NCT03863626

Study name	EDEMA trial
Methods	Randomised controlled trial

NCT03863626 (Continued)

Participants	Adults with chronic heart failure prior diagnosis, based on signs and symptoms of heart failure, presenting with acute decompensation as judged by the physician to require hospitalisation for intravenous diuretics
Interventions	Continuous infusion vs bolus injection of intravenous furosemide Metolazone will be added for sequential blockade in the intravenous injections arm qualifying diuretic resistance, there will be second level of randomisation into time-adjusted versus random time metolazone administration.
Outcomes	Primary outcome: time to improvement of New York Heart Association Class, diuretic efficiency (urinary output per 40 mg of furosemide administered as bolus or continuous infusion). Secondary outcome: effectiveness of sequential nephron block, evaluating superiority of time-adjusted metolazone compared to given at random in overcoming resistance to intravenous frusemide (only in intravenous bolus arm).
Starting date	31 March 2019
Contact information	Ahmad Samir, MD Cairo University Tel: 00201002647275 Email: ahmad.samir@kasralainy.edu.eg
Notes	

Palazzuoli 2017

Study name	DIUR-AHF study
Methods	Multicentre randomised controlled trial
Participants	Adults meeting the diagnostic criteria for acute heart failure, independently of the systolic function, by exhibiting ≥ 1 symptom at rest among dyspnoea, orthopnoea, peripheral oedema, and major fatigue; and ≥ 2 clinical signs including rales, pulmonary congestion on chest X-ray (pulmonary oedema, pulmonary congestion, or pleural effusion), jugular vein dilation, and a third heart sound. An elevation in blood B-type natriuretic peptide ≥ 100 pg/mL is considered supportive for a diagnosis of acute heart failure.
Interventions	Quote: "At admission, patients with AHF in both the intermittent and continuous infusion arms will be administered with 80 mg/day of intravenous furosemide. If a patient has a good response, the initial dose will be continued; if a patient has a poor response, defined as diuretic efficiency < 0.2 kg/day for 40 mg of furosemide and a urine output < 1 L/day, the furosemide dosage will be doubled (160 mg/day). In case of continued diuretic resistance, defined as a poor response, the dosage will be raised to 250 mg/day."
Outcomes	Primary endpoints: 180-day cardiac death and rehospitalisation, evaluation of adverse events in relation to the rate of worsened renal function and to congestion signs after the infusion period during a 6-month follow-up period. Secondary endpoints: mean paired changes of renal function, of urine output comparing continuous vs intermittent administration; the evaluation of decongestion, B-type natriuretic peptide decrease, and worsened renal function occurrence in relation to diuretic efficiency; post hoc analy-

Palazzuoli 2017 (Continued)

sis of worsening of renal function, congestion signs, inotropic infusion, and length of hospital stay (days) should be performed with respect to the specific dose groups (high vs low).

Starting date	December 2015
Contact information	Dr Alberto Palazzuoli Department of Internal Medicine, Cardiovascular Diseases Unit, S. Maria alle Scotte Hospital, University of Siena, Viale Bracci, 53100 Siena, Italy Tel: +39577585363; +39577585466 Fax: +39577233480 Email: palazzuoli2@unisi.it
Notes	

RISK OF BIAS

Legend:  Low risk of bias  High risk of bias  Some concerns

Risk of bias for analysis 1.1 Net weight loss (kg)

Study	Bias					Overall
	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	
Allen 2010						
Felker 2011						
Sager 2020						
Yayla 2015						
Zheng 2021						

Risk of bias for analysis 1.2 All-cause mortality

Study	Bias					Overall
	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	
Allen 2010	~	✓	✓	✓	~	~
Felker 2011	✓	✓	✓	✓	✓	✓
Sager 2020	✗	~	✓	✓	~	✗
Shah 2014	~	~	✓	✓	~	~
Zheng 2021	~	~	✗	✓	~	✗

Risk of bias for analysis 1.3 Length of hospital stay (days)

Study	Bias					Overall
	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	
Allen 2010	~	✓	✓	✓	~	~
Shah 2014	~	✓	✓	✓	~	~
Yayla 2015	✓	✓	✓	✓	~	~
Zheng 2021	~	~	✗	✓	~	✗

Risk of bias for analysis 1.4 Readmission to hospital following discharge for the longest available follow-up stated in the trials

Study	Bias					Overall
	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	
Felker 2011	✓	✓	✓	✓	✓	✓
Sager 2020	✗	~	✓	✓	~	✗

Bias						
Study	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	Overall
Shah 2014	~	~	✓	✓	~	~

Risk of bias for analysis 1.5 Acute kidney injury (adverse event)

Bias						
Study	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	Overall
Felker 2011	✓	✓	✓	✓	✓	✓
Llorens 2014	✓	✓	✓	✓	✓	✓
Zheng 2021	~	~	✗	✓	~	✗

Risk of bias for analysis 2.1 Net weight loss (kg)

Bias						
Study	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	Overall
Allen 2010	~	✓	✓	✓	~	~
Felker 2011	✓	✓	✓	✓	✓	✓
Yayla 2015	✓	✓	✓	✓	~	~

Risk of bias for analysis 2.2 All-cause mortality

Study	Bias					Overall
	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	
Allen 2010						
Felker 2011						
Shah 2014						

Risk of bias for analysis 2.3 Length of hospital stay (days)

Study	Bias					Overall
	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	
Allen 2010						
Shah 2014						
Yayla 2015						

Risk of bias for analysis 2.4 Readmission to hospital following discharge for the longest available follow-up stated in the trials

Study	Bias					Overall
	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	
Felker 2011						
Shah 2014						

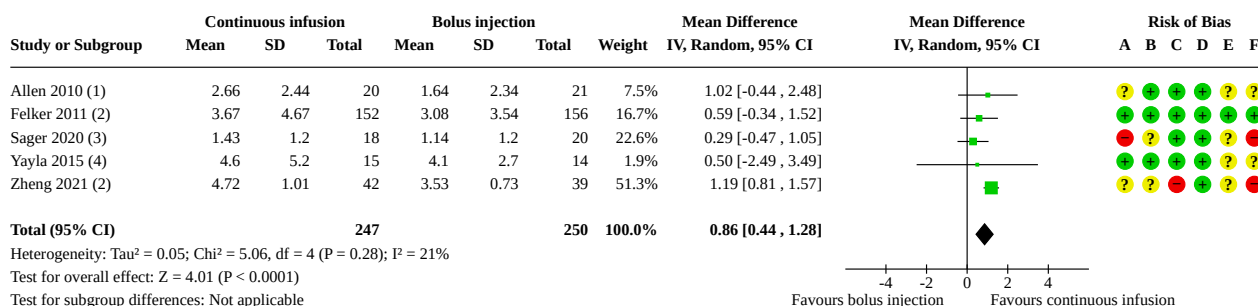
Risk of bias for analysis 2.5 Acute kidney injury (adverse event)

Study	Bias					Overall
	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	
Felker 2011	✓	✓	✓	✓	✓	✓
Llorens 2014	✓	✓	✓	✓	✓	✓

DATA AND ANALYSES

Comparison 1. Meta-analysis continuous infusion versus bolus injection

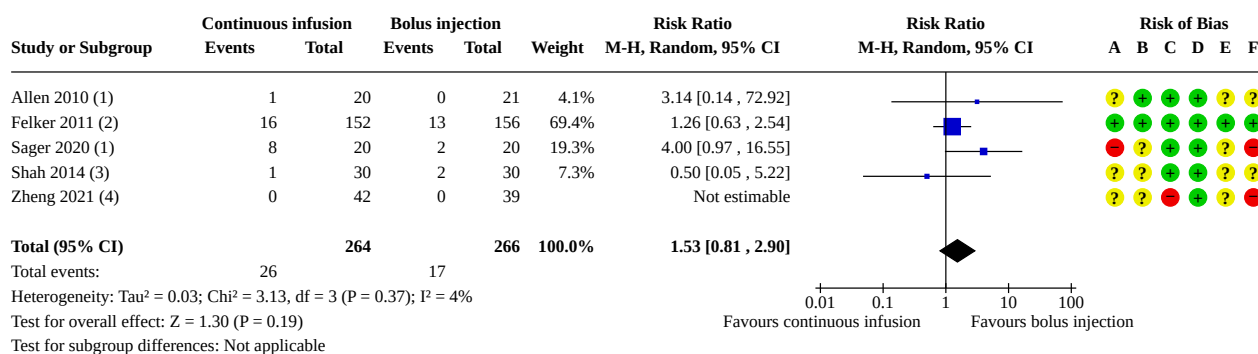
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Net weight loss (kg)	5	497	Mean Difference (IV, Random, 95% CI)	0.86 [0.44, 1.28]
1.2 All-cause mortality	5	530	Risk Ratio (M-H, Random, 95% CI)	1.53 [0.81, 2.90]
1.3 Length of hospital stay (days)	4	211	Mean Difference (IV, Random, 95% CI)	-1.10 [-4.84, 2.64]
1.4 Readmission to hospital following discharge for the longest available follow-up stated in the trials	3	400	Risk Ratio (IV, Fixed, 95% CI)	0.85 [0.63, 1.16]
1.5 Acute kidney injury (adverse event)	3	491	Risk Ratio (M-H, Random, 95% CI)	1.02 [0.70, 1.49]

**Analysis 1.1. Comparison 1: Meta-analysis continuous infusion
versus bolus injection, Outcome 1: Net weight loss (kg)****Footnotes**

- (1) Until hospital day 3 or discharge, whichever came first.
(2) At 72 hours.
(3) Until discharge.
(4) Until decompensation.

Risk of bias legend

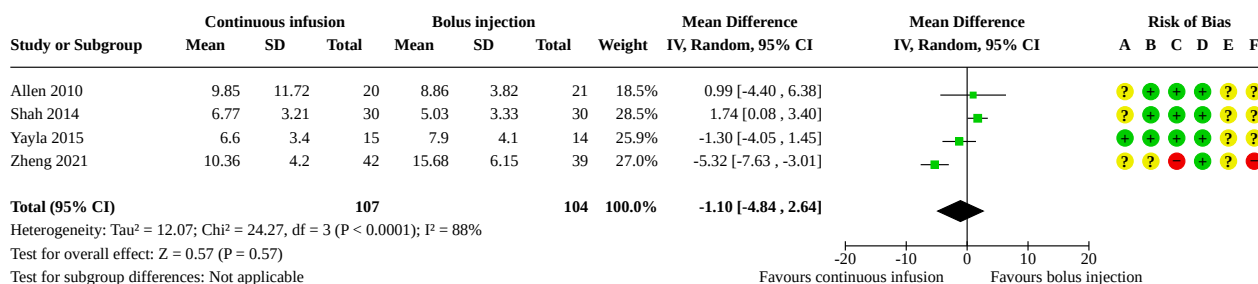
- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

**Analysis 1.2. Comparison 1: Meta-analysis continuous
infusion versus bolus injection, Outcome 2: All-cause mortality****Footnotes**

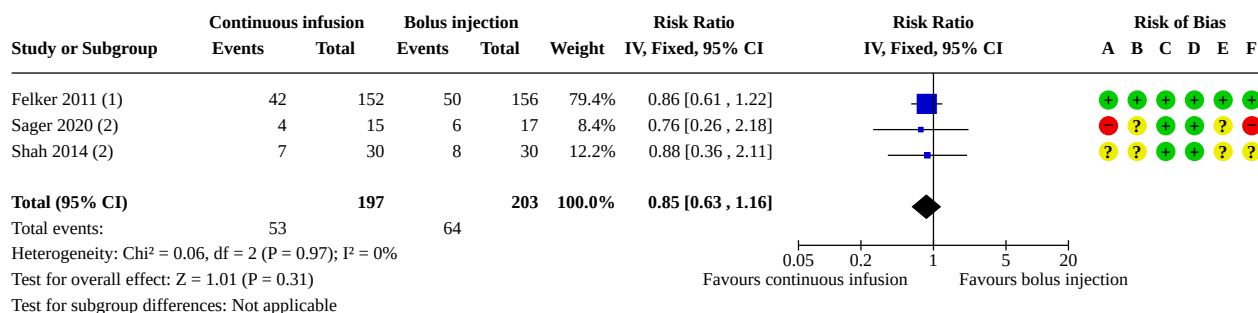
- (1) 90 days.
(2) 60 days.
(3) 30 days.
(4) During hospitalisation.

Risk of bias legend

- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

**Analysis 1.3. Comparison 1: Meta-analysis continuous infusion
versus bolus injection, Outcome 3: Length of hospital stay (days)****Risk of bias legend**

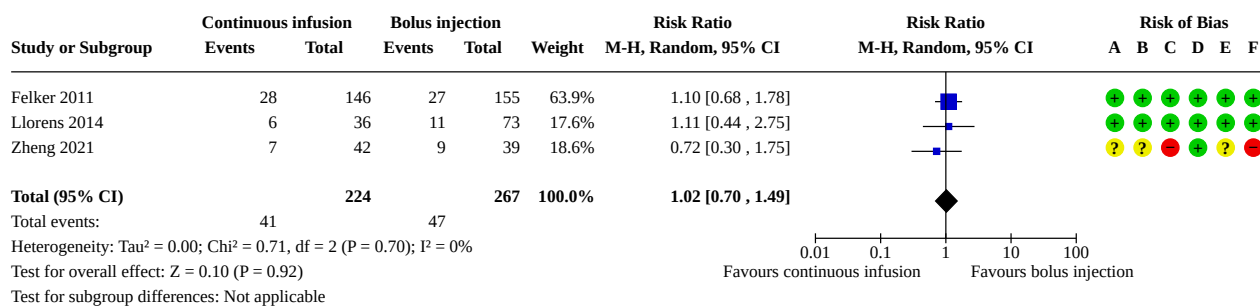
- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

**Analysis 1.4. Comparison 1: Meta-analysis continuous infusion versus bolus injection, Outcome 4:
Readmission to hospital following discharge for the longest available follow-up stated in the trials****Footnotes**

- (1) 60 days.
(2) 30 days.

Risk of bias legend

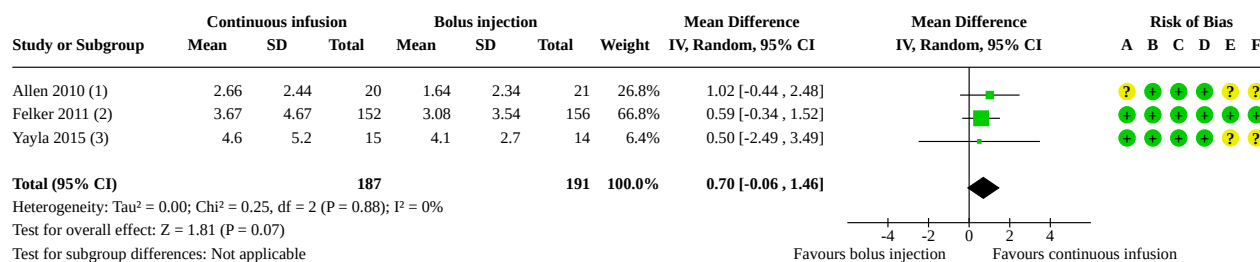
- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Analysis 1.5. Comparison 1: Meta-analysis continuous infusion versus bolus injection, Outcome 5: Acute kidney injury (adverse event)**Risk of bias legend**

- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Comparison 2. Sensitivity analysis – exclusion of trials with high risk of bias

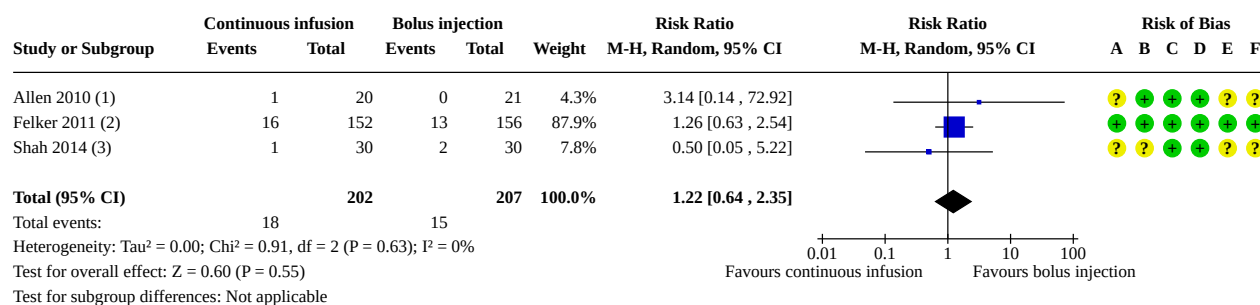
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Net weight loss (kg)	3	378	Mean Difference (IV, Random, 95% CI)	0.70 [-0.06, 1.46]
2.2 All-cause mortality	3	409	Risk Ratio (M-H, Random, 95% CI)	1.22 [0.64, 2.35]
2.3 Length of hospital stay (days)	3	130	Mean Difference (IV, Random, 95% CI)	0.61 [-1.53, 2.75]
2.4 Readmission to hospital following discharge for the longest available follow-up stated in the trials	2	368	Risk Ratio (IV, Fixed, 95% CI)	0.86 [0.63, 1.19]
2.5 Acute kidney injury (adverse event)	2	410	Risk Ratio (M-H, Random, 95% CI)	1.10 [0.72, 1.68]

**Analysis 2.1. Comparison 2: Sensitivity analysis – exclusion
of trials with high risk of bias, Outcome 1: Net weight loss (kg)****Footnotes**

- (1) Until hospital day 3 or discharge, whichever came first.
(2) At 72 hours.
(3) Until decompensation.

Risk of bias legend

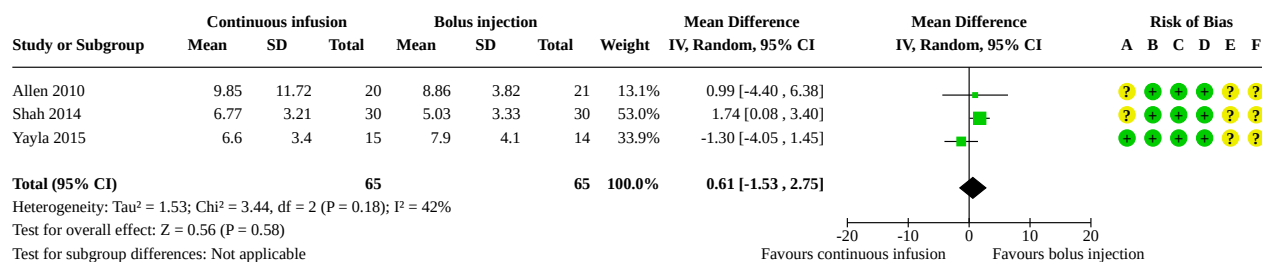
- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

**Analysis 2.2. Comparison 2: Sensitivity analysis – exclusion
of trials with high risk of bias, Outcome 2: All-cause mortality****Footnotes**

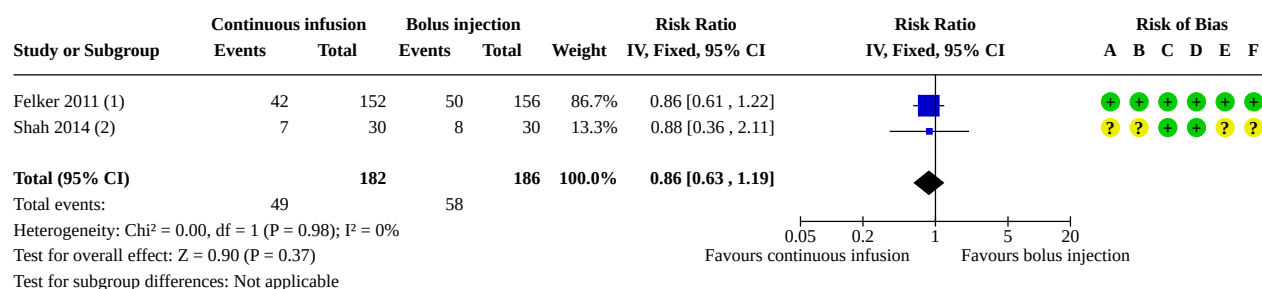
- (1) 90 days.
(2) 60 days.
(3) 30 days.

Risk of bias legend

- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Analysis 2.3. Comparison 2: Sensitivity analysis – exclusion of trials with high risk of bias, Outcome 3: Length of hospital stay (days)**Risk of bias legend**

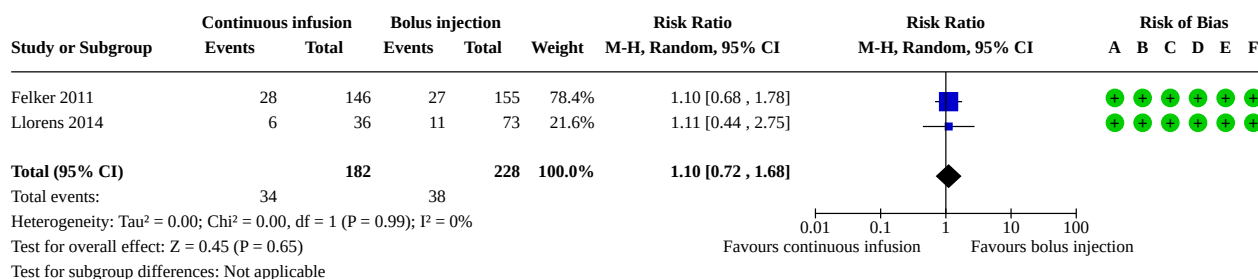
- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Analysis 2.4. Comparison 2: Sensitivity analysis – exclusion of trials with high risk of bias, Outcome 4: Readmission to hospital following discharge for the longest available follow-up stated in the trials**Footnotes**

- (1) 60 days.
(2) 30 days.

Risk of bias legend

- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Analysis 2.5. Comparison 2: Sensitivity analysis – exclusion of trials with high risk of bias, Outcome 5: Acute kidney injury (adverse event)**Risk of bias legend**

- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

ADDITIONAL TABLES**Table 1. Accumulative net urinary output volume**

Study ID	Net urinary output in continuous infusion (millilitre, mean, n)	Net urinary output in bolus injection (millilitre, mean, n)	Duration of observation	P value
Felker 2011	4249 (SD 3104), n = 152	4237 (SD 3208), n = 156	72 hours	0.89
Zheng 2021	5145.98 (SD 621.37), n = 42	3755.95 (SD 456.93), n = 39	72 hours	0.01

n: number of participants; SD: standard deviation.

APPENDICES**Appendix 1. Initial search strategies for bibliographic databases****Cochrane CENTRAL**

- #1 MeSH descriptor: [Heart Failure] explode all trees
#2 ((heart* or cardiac* or myocard*) NEAR/2 (fail* or insuff*))
#3 (heart* NEAR/2 decomp*)
#4 (HF or CHF or ADHF or HFReF or HFpEF or HFmrEF)
#5 #1 or #2 or #3 or #4
#6 MeSH descriptor: [Sodium Potassium Chloride Symporter Inhibitors] explode all trees
#7 (sodium potassium chloride symporter NEAR/3 inhibit*)
#8 loop diuretic*
#9 (high ceiling NEAR/3 diuretic*)

#10 (furosemid* or frusemid* or fursemid* or furantral)

#11 (torsemide* or torasemid*)

#12 (bumetanide* or bumethanid*)

#13 (etacrynic acid* or ethacrynic acid*)

#14 #6 or #7 or #8 or #9 or #10 or #11 or #12 or #13

#15 #5 and #14

MEDLINE Ovid

1 exp Heart Failure/ (130790)

2 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff*)).tw. (200967)

3 (heart* adj2 decomp*).tw. (4678)

4 (HF or CHF or ADHF or HFrEF or HFpEF or HFmrEF).tw. (67581)

5 1 or 2 or 3 or 4 (263131)

6 exp Sodium Potassium Chloride Symporter Inhibitors/ (14281)

7 (sodium potassium chloride symporter adj3 inhibit*).tw. (1)

8 loop diuretic*.tw. (3211)

9 (high ceiling adj3 diuretic*).tw. (102)

10 (furosemid* or frusemid* or fursemid* or furantral).tw. (13546)

11 (torsemide* or torasemid*).tw. (466)

12 (bumetanide* or bumethanid*).tw. (3107)

13 (etacrynic acid* or ethacrynic acid*).tw. (2006)

14 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 (23889)

15 5 and 14 (3458)

16 randomized controlled trial.pt. (542905)

17 controlled clinical trial.pt. (94381)

18 randomized.ab. (533051)

19 placebo.ab. (221207)

20 clinical trials as topic.sh. (197266)

21 randomly.ab. (365421)

22 trial.ti. (247101)

23 16 or 17 or 18 or 19 or 20 or 21 or 22 (1392382)

24 exp animals/ not humans.sh. (4883379)

25 23 not 24 (1281380)

26 15 and 25 (703)

Embase Ovid

1 exp heart failure/ (535282)

- 2 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff*)).tw. (321222)
- 3 (heart* adj2 decomp*).tw. (9502)
- 4 (HF or CHF or ADHF or HFrEF or HFpEF or HFmrEF).tw. (120018)
- 5 1 or 2 or 3 or 4 (629532)
- 6 exp loop diuretic agent/ (69349)
- 7 (sodium potassium chloride symporter adj3 inhibit*).tw. (0)
- 8 loop diuretic*.tw. (5206)
- 9 (high ceiling adj3 diuretic*).tw. (126)
- 10 (furosemid* or frusemid* or fursemid* or furantral).tw. (17587)
- 11 (torsemide* or torasemid*).tw. (835)
- 12 (bumetanide* or bumethanid*).tw. (3524)
- 13 (etacrynic acid* or ethacrynic acid*).tw. (1400)
- 14 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 (72572)
- 15 5 and 14 (18759)
- 16 random\$.tw. (1686447)
- 17 factorial\$.tw. (41221)
- 18 crossover\$.tw. (79828)
- 19 cross over\$.tw. (33388)
- 20 cross-over\$.tw. (33388)
- 21 placebo\$.tw. (324597)
- 22 (doubl\$ adj blind\$).tw. (215777)
- 23 (singl\$ adj blind\$).tw. (27179)
- 24 assign\$.tw. (426025)
- 25 allocat\$.tw. (169572)
- 26 volunteer\$.tw. (264538)
- 27 crossover procedure/ (67870)
- 28 double blind procedure/ (184213)
- 29 randomized controlled trial/ (669738)
- 30 single blind procedure/ (43577)
- 31 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 (2511615)
- 32 (animal/ or nonhuman/) not human/ (5849470)
- 33 31 not 32 (2227281)
- 34 15 and 33 (2304)
- 35 limit 34 to embase (1813)

CPCI-S

15 #14 AND #13

14 TS=(random* or blind* or allocat* or assign* or trial* or placebo* or crossover* or cross-over*)

13 #12 AND #4

12 #11 OR #10 OR #9 OR #8 OR #7 OR #6 OR #5

11 TS=(etacrynic acid* or ethacrynic acid*)

10 TS=(bumetanide* or bumethanid*)

9 TS=(torsemide* or torasemid*)

8 TS=(furosemid* or frusemid* or fursemid* or furantral)

7 TS=(high ceiling NEAR/3 diuretic*)

6 TS=loop diuretic*

5 TS=(sodium potassium chloride symporter)

4 #3 OR #2 OR #1

3 TS=(HF or CHF or ADHF or HFrEF or HFpEF or HFmrEF)

2 TS=(heart* NEAR/2 decomp*)

1 TS=((heart* or cardiac* or myocard*) NEAR/2 (fail* or insuff*))

ClinicalTrials.gov

Condition or disease: Heart Failure

Study type: Interventional Studies (Clinical Trials)

Intervention/treatment: Loop diuretic

WHO ICTRP

Condition: "heart failure"

Intervention: "loop diuretic"

EU clinical trial register

"heart failure" AND "loop diuretic"

Appendix 2. Top-up search strategies for bibliographic databases**MEDLINE**

1 exp heart failure/ 152132

2 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff)).mp. 277984

3 (heart* adj2 decomp*).mp. 5749

4 (HF or CHF or ADHF or HFrEF or HFpEF or HFmrEF).mp. 83922

5 1 or 2 or 3 or 4 311372

6 exp Sodium Potassium Chloride Symporter Inhibitors/ 14849

7 (sodium potassium chloride symporter adj3 inhibit*).mp. 1334

8 loop diuretic*.mp. 3725

9 (high ceiling adj3 diuretic*).mp. 104
10 (furosemid* or frusemid* or fursemid* or furantral).mp. 18309
11 (torsemide* or torasemid*).mp. 597
12 (bumetanide* or bumethanid*).mp. 3566
13 (etacrynic acid* or ethacrynic acid*).mp. 2906
14 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 25741
15 5 and 14 4003
16 randomized controlled trial.pt. 609693
17 controlled clinical trial.pt. 95571
18 randomized.ab. 632112
19 placebo.ab. 245071
20 clinical trials as topic.sh. 201817
21 randomly.ab. 425105
22 trial.ti. 301346
23 16 or 17 or 18 or 19 or 20 or 21 or 22 1574589
24 exp animals/ not humans.sh. 5200495
25 23 not 24 1449295
26 15 and 25 797
27 limit 26 to yr="2021 -Current" 118

Embase

1 exp heart failure/ 661979
2 ((heart* or cardiac* or myocard*) adj2 (fail* or insuff*)).mp. 538114
3 (heart* adj2 decomp*).mp. 11820
4 (HF or CHF or ADHF or HFrEF or HFpEF or HFmrEF).mp. 150793
5 1 or 2 or 3 or 4 783241
6 exp loop diuretic agent/ 86261
7 (sodium potassium chloride symporter adj3 inhibit*).mp. 8
8 loop diuretic*.mp. 15085
9 (high ceiling adj3 diuretic*).mp. 150
10 (furosemid* or frusemid* or fursemid* or furantral).mp. 69929
11 (torsemide* or torasemid*).mp. 3547
12 (bumetanide* or bumethanid*).mp. 7759
13 (etacrynic acid* or ethacrynic acid*).mp. 4952
14 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 90154
15 5 and 14 24345

16 random\$.mp. 2314126

17 factorial\$.mp. 75629

18 crossover\$.mp. 121549

19 cross over\$.mp. 38759

20 cross-over\$.mp. 38759

21 placebo\$.mp. 534219

22 (doubl\$ adj blind\$).mp. 307317

23 (singl\$ adj blind\$).mp. 66607

24 assign\$.mp. 507057

25 allocat\$.mp. 227328

26 volunteer\$.mp. 310519

27 crossover procedure/ 77117

28 double blind procedure/ 216408

29 randomized controlled trial/ 809441

30 single blind procedure/ 53784

31 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 3315745

32 animal/ 1650914

33 nonhuman/ 7634668

34 32 or 33 9267321

35 human/ 26069596

36 34 not 35 6588919

37 31 not 36 2965549

38 15 and 37 3814

39 limit 38 to embase 3244

40 limit 39 to yr="2021 -Current" 664

COCHRANE CENTRAL

#1 MeSH descriptor: [Heart Failure] explode all trees

#2 ((heart* OR cardiac* OR myocard*) NEAR/2 (fail* or insuff*)) with Publication Year from 2021 to 2024, in Trials

#3 (heart* NEAR/2 decomp*) with Publication Year from 2021 to 2024, in Trials

#4 (HF OR CHF OR ADHF OR HFrEF OR HFpEF OR HFmrEF) with Publication Year from 2021 to 2024, in Trials

#5 #1 OR #2 OR #3 OR #4 with Cochrane Library publication date Between Sep 2021 and Feb 2024

#6 MeSH descriptor: [Sodium Potassium Chloride Symporter Inhibitors] explode all trees

#7 (sodium potassium chloride symporter NEAR/3 inhibit*) with Publication Year from 2021 to 2024, in Trials

#8 loop diuretic* with Publication Year from 2021 to 2024, in Trials

#9 (high ceiling NEAR/3 diuretic*) with Publication Year from 2021 to 2024, in Trials

#10 (furosemid* OR frusemid* OR fursemid* OR furantral) with Publication Year from 2021 to 2024, in Trials

#11 (torsemide* OR torasemid*) with Publication Year from 2021 to 2024, in Trials

#12 (bumetanide* OR bumethanid*) with Publication Year from 2021 to 2024, in Trials

#13 (etacrynic acid* OR ethacrynic acid*) with Publication Year from 2021 to 2024, in Trials

#14 #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 with Publication Year from 2021 to 2024, in Trials

#15 #5 AND #14 with Publication Year from 2021 to 2024, in Trials

Continuous infusion versus bolus injection of loop diuretics for acute heart failure (Review)

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CPCI-S

15 #14 AND #13 Publication date: 2021-09-01 to 2024-02-29

14 TS=(random* or blind* or allocat* or assign* or trial* or placebo* or crossover* or cross-over*)

13 #12 AND #4

12 #11 OR #10 OR #9 OR #8 OR #7 OR #6 OR #5

11 TS=(etacrynic acid* or ethacrynic acid*)

10 TS=(bumetanide* or bumethanid*)

9 TS=(torsemide* or torasemid*)

8 TS=(furosemid* or frusemid* or fursemid* or furantral)

7 TS=(high ceiling NEAR/3 diuretic*)

6 TS=loop diuretic*

5 TS=(sodium potassium chloride symporter)

4 #3 OR #2 OR #1

3 TS=(HF or CHF or ADHF or HFrEF or HFpEF or HFmrEF)

2 TS=(heart* NEAR/2 decomp*)

1 TS=((heart* or cardiac* or myocard*) NEAR/2 (fail* or insuff*))

ClinicalTrials.gov*

Condition or disease: Heart Failure

Study type: Interventional Studies (Clinical Trials)

Intervention/treatment: Loop diuretic

ICTRP

Condition: "heart failure"

Intervention: "loop diuretic"

Date of registration is between 01/09/2021 and 29/02/2024

EU clinical trial register*

"heart failure" AND "loop diuretic"

* Search range limited to run from first/original search to top-up search dates

HISTORY

Protocol first published: Issue 8, 2021

CONTRIBUTIONS OF AUTHORS

DR: data accuracy review, analysis of results, top-up database searches and screening, writing and completion of manuscript and project.

JZ: title and abstract screening, full-text review, risk of bias assessment, data extraction, analysis of the results, top-up database search results screening, drafting of the manuscript.

EF: title and abstract screening, full-text review.

AT: title and abstract screening, full-text review.

SCC: data extraction, risk of bias assessment.

RF: title and abstract screening.

CZ: risk of bias assessment.

MI: title and abstract screening, full-text review.

SA: full-text review.

TSL: full-text review.

JB: full-text review, data extraction, risk of bias assessment.

RH: proofreading and revision with methodological and statistical input.

GL: conception of research idea, organisation of the team, supervision of the research process, revision of the manuscript.

RS: revision of manuscript with clinical input.

DECLARATIONS OF INTEREST

DR declares no conflicts of interest related to this project. Speaker fee received from Novartis previously.

JZ: none.

EF: none.

AT: none.

SCC: none.

RF: none.

CZ: none.

MI: none.

SA: none.

TSL: none.

JB: none.

RH declares a financial – non-personal, non-specific – interest, having delivered educational workshops on health economics, medicines management and healthy technology assessment for cancer specialists supported by unrestricted sponsorship by the pharmaceutical industry and an industry association (March 2019). No fees received personally. Not specific to the topic of the review. This research is supported by NIHR Systematic Reviews Cochrane Programme (16/114/26) and NIHR TAR (2016–2021) funding.

GL declares being a consultant for Bayer/Janssen, BMS/Pfizer, Medtronic, Boehringer Ingelheim, Novartis, Verseon and Daiichi-Sankyo; speaker for Bayer, BMS/Pfizer, Medtronic, Boehringer Ingelheim, and Daiichi-Sankyo. No fees are received personally.

RS declares no direct conflicts of interest related to this project. Speaker fees received from Novartis, Sanofi, BMS/Pfizer, AstraZeneca. Research Grant from Biotronik previously. Travel grant from AstraZeneca.

SOURCES OF SUPPORT

Internal sources

- None, Other

No internal sources of support was used for this review.

External sources

- NIHR, UK

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DIFFERENCES BETWEEN PROTOCOL AND REVIEW

Following a review of the literature, all included studies have reported changes in weight using net weight loss (baseline bodyweight minus bodyweight following treatment in kilograms (kg)). No study reported weight loss as a percentage change of bodyweight as we had initially proposed in our protocol ([Zhang 2021](#)). Following discussion with methodological and clinical heart failure experts in our team, we reached a consensus to use net weight loss (kg) in our analysis.

Given that two out of the seven trials included in our analysis exhibited overall high risk of bias and an additional three trials demonstrated some concerns overall, it would neither be practical nor informative to exclude all of them from our analysis. While our original intention was to exclusively include studies with a low risk of bias, the limited number of studies and participants in this review would significantly diminish its clinical applicability value. Consequently, we have opted to only exclude trials with overall high risk of bias, while ensuring a thorough evaluation and representation of our certainty of evidence throughout the review process.

In our protocol, we planned to conduct subgroup analyses; however, due to a lack of available subgroup data in the trials included in the review, subgroup analysis could not be performed ([Zhang 2021](#)).

In accordance with the Cochrane MECIR standard C37, we conducted an additional top-up literature review in February 2024 to ensure that our search results were within six months of publication. Based on guidance from a Cochrane Information Specialist, we refined our search term extension from the protocol previously published. Amendment in [Appendix 2](#).

As outlined in our protocol, our original intention was to analyse our results using a funnel plot ([Zhang 2021](#)). However, due to the insufficient number of studies meeting our inclusion criteria (fewer than 10), we did not create a funnel plot as planned.

NOTES

This review supersedes a previous review publication ([Salvador 2005](#)).

INDEX TERMS

Medical Subject Headings (MeSH)

Acute Disease; Bias; Cause of Death; *Heart Failure [drug therapy]; Infusions, Intravenous; Injections, Intravenous; Length of Stay; Randomized Controlled Trials as Topic; *Sodium Potassium Chloride Symporter Inhibitors [administration & dosage] [adverse effects]

MeSH check words

Adult; Aged; Humans